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Price Discovery and Efficiency in Uniswap Liquidity Pools

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ABSTRACT

Using almost three years of minute-level data, we show that the efficiency of Uniswap v3 is much improved relative to v2, and some v3 pools are approaching or even exceeding Bitstamp in terms of price discovery ability. Regression results suggest that the channels of influence for these improvements are an increase in informed liquidity providers and swappers, who nevertheless prefer to use centralized exchanges when markets become volatile. At such times, there are relatively more uninformed speculative traders in all Uniswap pools, and cross-exchange arbitrage activities become more prevalent. However, these decrease the price discovery ability of Uniswap pools relative to Coinbase and Bitstamp. The informed traders that remain on Uniswap during periods of high uncertainty tend to switch to the higher-fee v3 pools to compensate for efficiency loss, or to v2 pools where there is sufficient liquidity to complete large trades.

JEL Classification: G14, D47

1 | Introduction

Crypto assets have very special trading features. There are two completely different market making processes. On centralized exchanges (CEXs), crypto prices are formed by standard limit order book matching system, but decentralized exchanges (DEXs) normally use a novel type of automatically managed liquidity pool, governed by smart contracts, to set price via an automated market making design (AMM), which enables direct peer-to-peer crypto transactions.

There are hundreds of CEXs, and all but a handful are unregulated by market authorities. For this reason, they can—and do—operate very differently from traditional exchanges for securities, commodities, or derivatives. There is little or no independent risk management oversight or even attempts to separate activities on most crypto CEXs.[1] The recent collapse of FTX is a good illustration of the problems. The event was triggered by a report from CoinDesk in early November 2022, which pointed out the potential leverage and solvency concerns regarding Alameda Research, an FTX-affiliated trading firm. Immediately, FTX lost billions in its valuation, followed by a liquidity crisis, eventually leading to its bankruptcy. Because activities on CEXs are highly anonymous and difficult to trace, price and volume manipulation are common (Cong et al. 2023; Gandal et al. 2018).[2] Most leading crypto CEXs apply know-your-customer rules and take custody of traders' assets to facilitate order matching. Hence, both the funds and private information of their clients are prone to intermediary defaults, privacy leaks, or blockchain wallet hackers. Examples of such incidents are common.[3]

The rising popularity of DEXs stems from all these concerns. Although the trading volume is still less than that of CEXs,

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DEXs resolve many of the challenges currently faced by CEXs. Indeed, the completely different order matching and trade management procedures have many implications for operational management. Because every transaction must be recorded on a blockchain, trading volume is limited by the speed of the blockchain. Most pools use the Ethereum chain, which currently adds about one block every 12 seconds (see Etherscan), but other chains or layer 2 solutions are possible. The leading DEX Uniswap currently allows pools to employ Arbitum, Base, BNB Chain, Celo, Optimism, Polygon, and Ethereum. Because of the speed limitation associated with blockchain transparency, it is difficult for DEX market participants to engage in wash trading (volume inflation) because gas fees must be paid for each transaction (Han et al. 2023).[4] Also, unlike a CEX, there is no need for custody on a DEX, where tokens remain in users' wallets and are never stored by any intermediary. Consequently, the security and privacy issues of centralized control are eliminated, making transactions more transparent and safe.[5] Indeed, Han et al. (2023) show that DEXs can gain investors' trust because of their ability to aggregate the consensus of a token's value when it is not feasible for the token's trading structure to be underwritten by a credible central party.

As of December 2024, the total value locked (TVL) in decentralized finance was about 140 billion USD, with around 30 billion locked in DEXs.[6] Figure 1 (left panel) shows that the monthly spot trading volume on DEXs has been growing to about 11% of the volume on CEXs. The right panel shows a recent proliferation of DEXs, with about 35% of current trading occurring on Uniswap. The major DEXs also serve as the price oracle for decentralized applications, such as the popular peer-to-peer lending platforms AAVE and Compound (Adams et al. 2022). Indeed, DEXs are rapidly becoming one of the central pillars of the decentralized finance ecosystem (Caldarelli and Ellul 2021; Dave et al. 2021). It is therefore important for academics and practitioners to understand their microstructure, as we do in this paper.

Information dissemination of DEXs is still relatively slow, and as trading volume grows, this could be severely hindered by the limited speed of the blockchain consensus mechanisms. So, as operational improvements in AMM protocols are made to increase trading volume and attract more sophisticated traders, does a DEX become more or less efficient? Could the price actually be slower to respond, due to blockchain congestion? We answer this question by investigating Uniswap's price discovery ability and market efficiency. As well as comparing the main centralized and decentralized markets for ether (ETH) spot trades, namely the USDC-ETH Uniswap v2 and v3 markets and the USD-ETH Coinbase and Bitstamp markets, we also compare another 10 matching token pairs that are traded on these platforms.

Our main conclusions are as follows: (1) Uniswap markets have lower levels of market efficiency compared with Coinbase and Bitstamp; they play the role of "shock receiver" so their global market influence is limited; (2) Informed traders prefer to use centralized exchanges when markets become volatile. However, trading volume increases during volatile periods, so more speculators must be trading in the Uniswap pools during periods of high uncertainty. The basis between Uniswap and Coinbase or Bitstamp also has a predominately negative effect on price discovery. Since this basis tends to widen during volatile periods, fewer informed traders but a greater number of uninformed traders are using Uniswap, e.g. for cross-exchange arbitrage, at these times; (3) The Uniswap v3 upgrade has much improved price discovery ability relative to Uniswap v2, and the price discovery ability of some Uniswap v3 pools can now even exceed that of Bistamp; (4) Considering pools with different fee structures, large trades decrease price discovery ability of the v3 (005) pools but increase price discovery of the larger v3 (030) and v2 pools. Hence, informed traders choose to make large trades on higher-fee pools where there should be sufficient liquidity to do the trade without too much slippage; (5) Regarding the inferior efficiency of high-fee pools, when markets are highly volatile the v3 liquidity provider needs to either (a) widen their supply range, in which case the capital efficiency decreases and the pool becomes more like the v2 pools, or (b) change a narrow supply range more frequently, thus incurring gas fees. Either action will decrease market efficiency.

In the following: Section 2 sets our work in the context of the relevant literature; Section 3 describes the technological improvements of Uniswap v3 compared with Uniswap v2 as they relate to (a) capital efficiency and (b) market efficiency; Section 4 describes our data and lists the statistical tests employed; Section 5 reports our empirical findings; and Section 6 concludes. An Appendix describes the statistical tests and presents some supplementary empirical results. Our data analysis is built using Python and R; all scripts are available upon request.

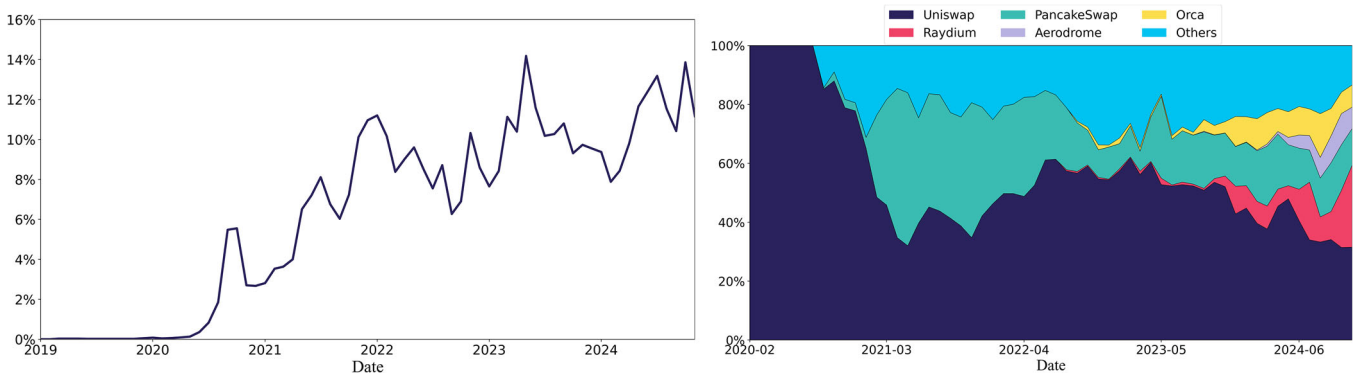

FIGURE 1 | Spot trading volume on DEXs. Total DEXs monthly trading volume relative to that on CEXs (left panel; *source:* The Block) and monthly trading volume shares between major DEXs (right panel; *source:* The Block).

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2 | Positioning in the Literature

Computer science is by far the most active of all the disciplines focusing research on blockchain trading. However, whenever an institution deploys its own capital for any length of time, it must consider whether this capital is allocated in an optimal manner. So this paper examines the development of the main decentralized market for some major token pairs in terms of price discovery and efficiency. These concepts have been applied to traditional markets for many decades, using well-known models and metrics. They measure how quickly new information is incorporated into asset prices, and how prices are transmitted through different instruments on the same underlying assets. More recent studies examine crypto markets, which have a much more interesting microstructure because the same product can be traded in many different venues. Several recent papers have focused on the efficiency of CEXs, which, not surprisingly, have become more efficient over time—see Krückeberg and Scholz (2020), Alexander et al. (2020), Alexander et al. (2022), and many others since. But relatively few studies have looked into the market efficiency of DEXs. Angeris et al. (2021) show that prices in the Uniswap market closely track the reference prices in centralized markets. Barbon and Ranaldo (2021) show that CEXs have better market quality than DEXs (in terms of lower transaction costs and fewer deviations from the no-arbitrage condition), but they argue that improvements in Uniswap v3 smart contracts can compensate for this lesser market quality, to some extent. We also conduct a comparative analysis of Uniswap markets relative to centralized exchanges, under different smart contracts, and we go further—to examine the drivers of the greatly increased efficiency more recently found in v3. We also use a vector error correction model (VECM) to examine how information is transmitted between CEXs and DEXs, and we calculate spillover effects to gain a clearer depiction of this transmission process. Lo and Medda (2021) only examine the efficiency of Uniswap v2, and they employ hourly data, which is not of sufficiently high frequency for a VECM-based price discovery analysis. We use minute-level data to examine the efficiency of both Uniswap v2 and v3 markets, and compare the results with those from centralized exchanges during the same time period.

Other relevant studies in the financial markets literature include: Aoyagi (2020), Loesch et al. (2021) and Heimbach et al. (2022b), who analyze the profits and losses of AMMs; Capponi and Jia (2021), Neuder et al. (2021) and Lehar and Parlour (2024), who analyze market-making strategies on DEXs; and Aigner and Dhaliwal (2021) and Deng et al. (2023), who investigate risk management on DEXs; Capponi et al. (2023), who study the emergence of private mining pools and its affect on price discovery; Lehar et al. (2023), who analyze the impact of gas fees and other transaction costs on liquidity provision strategies; and Han et al. (2023), who investigate investors' cross-market arbitrage activities based on price spreads between DEXs and CEXs and other effective information. Other studies more closely related to our work include: Hansson (2024), who analyses price discovery in constant product markets by demonstrating the impact of large, medium and small trades made by different types of traders of USDC-ETH on Uniswap v2; Entrop et al. (2020), who find that market uncertainty and trade characteristics (costs, volume and size) all contribute to price discovery of bitcoin in centralized spot and futures markets; and Chen and Yang (2024), who find that relative liquidity, investor structure and crypto hacking activities all contribute to the bitcoin futures market's price discovery process.

We extend these lines of research by investigating the effect of operational innovations—in terms of offering AMMs with a choice of liquidity provision range, and liquidity pools with different transactions costs—on the information content of market prices and the price discovery ability of a Uniswap pool in a global setting. We also analyze the effect of large trades and different market conditions as determinants of each Uniswap pool's market efficiency and price discovery ability, these dependent variables being measured using well-known models and metrics. Our paper differs from Hansson (2024) by demonstrating the inferior relative price discovery ability of Uniswap relative to major CEXs. In this sense, it complements Klein et al. (2023), who examine the USDC-ETH pools on Uniswap v3 versus major CEXs. They investigate the price impact of the flow of different types of orders, showing that by following the liquidity providers' activities on Uniswap, traders can place large limit orders on the CEXs, which create long-term price impacts.

Capponi et al. (2024) asks how information gets incorporated to asset prices on DEXs, showing that informed traders bid higher fees to execute their transactions and this may create relatively long-term price impacts; and in a similar vein, Mohan (2022) argues that the unique AMM mechanisms of DEXs could be a catalyst for a novel type of information transmission process, which in turn could impact CEX prices. By contrast, we do not ask *how* prices are formed in DEXs, and although it is potentially interesting to study the microstructure of the Uniswap market, in this paper, we investigate the role played by Uniswap in ether price discovery, instead of having prices that simply follow those formed on traditional CEX exchanges.

Another related strand of the finance literature looks at the trading volume and the liquidity provision and/or taking on DEXs. Park (2023) discusses the shortcomings of a deterministic liquidity invariance rule in AMM market smart contracts, including so-called sandwich attacks, and Adams and Liao (2022) highlight the advantages of Uniswap v3 over Uniswap v2 in terms of returns and liquidity. Our research complements these studies by comparing the informational inefficiency of both DEXs under different AMM mechanisms. Additionally, we reveal how improvements in smart contracts can increase both trading volume and market liquidity. In the same vein, a recent study by Hasbrouck et al. (2022) demonstrates that increasing trading fees can reduce the price impact on a DEX, also proposing a way to increase trading volume through smart contract design. For instance, the coming Uniswap v4 introduces Hooks, which bring greater composability and user flexibility while significantly reducing gas fees (Adams et al. 2023). This further innovation should attract greater trading volumes. Hasbrouck et al. (2025) derive an equilibrium level of liquidity provision within any given price interval and further demonstrate that providing liquidity for risky and risk-free assets (such as stablecoins and native tokens) is analogous to investing in a portfolio composed of both types of assets. Moving now to the other type of DEX users, Cartea et al. (2023a) focus on optimizing the

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large positions of a liquidity taker, developing trading strategies suitable for AMM markets with more or less efficiency. Our price discovery results also provide valuable insights for traders and investors to use when formulating their strategies, but instead of liquidity provision or taking, we uncover the potential for arbitrage opportunities between CEXs and DEXs. Like us, Cartea et al. (2023b) find that CEX prices always lead DEX markets, but we go further to look at the impact of information flows, the influence of smart contract design on efficiency, and the drivers of efficiency improvements.

3 | Uniswap Automated Market Making Mechanisms

Trading on Uniswap requires first selecting a liquidity pool, or *pool* for short. Every pool is linked to a token pair (X, Y), and the *pool price* P of the token X in terms of another token Y depends on the relative reserves of these two tokens in the pool, in a manner determined by the AMM protocols deployed. Uniswap also lists multiple pools for each pair, depending on: (i) the smart-contract blockchain used for validating transactions; and (ii) the fee paid by traders swapping tokens in the pool. In addition to trading fees, each user must pay the gas fees, tips to miners for validation, for transactions that depend on the blockchain chosen and the time-varying traffic activity levels on that blockchain.

Pools have two kinds of users, liquidity providers and liquidity takers. Liquidity providers, from henceforth simply *providers*, deposit simultaneously the two tokens in the ratio determined by the current pool price. In return for depositing tokens, providers receive a crypto asset called a *liquidity token,* which is minted via the DEX smart contract and could be used as collateral elsewhere in the Defi ecosystem. When a provider wishes to withdraw liquidity, he redeems the liquidity token, which is then burned by the smart contract, and the provider receives the original tokens in the ratio determined by the pool price at the time of burning. Liquidity takers pay a fee to providers to swap one token for another. We term this type of user a *swapper* or *taker* for short. Swapping alters the reserves ratio and thereby also the pool price.

Uniswap currently allows users to trade on two updates to its original (v1) smart contract protocol, namely v2 (launched on May 19, 2020) and v3 (launched on May 5, 2021).[7] They employ different versions of a Constant Product Market Maker (CPMM) mechanism. The general framework of CPMMs is described in the white paper of Adams et al. (2021) and analyzed by Angeris et al. (2021).[8] The fundamental principle of any CPMM is to maintain an invariant state in which the product of the quantities of tokens deposited by providers is held constant. Consider a pool with the quantity x of token X and the quantity y of token Y. The pool price P, namely the rate of exchange of token Y for token X, is the ratio of their reserves, that is, $P = y/x$. One illustrative example is taking token Y as the stable coin USDC as numéraire and token X as the more volatile crypto ETH. Thus, the relative price of X increases as the reserves of X decrease, and likewise for token Y.[9] Uniswap v2 requires that the product of the quantities of tokens X and Y should be constant, and set according to the so-called 'bonding curve', viz. the hyperbola:

$$xy = L^2, x, y \in (0, \infty), \tag{1}$$

where L is the *liquidity* supplied to the pool, which changes every time a provider makes or withdraws supplies of the two tokens, which must be done in the ratio determined by the current pool price. Uniswap v2 is designed to provide liquidity for any $P \in (0, \infty)$ rather than prices that trade in some anticipated finite range around the current price.

Now, suppose that the initial state of the v2 pool has x tokens of X and y tokens of Y, and the trader wants to swap token Y for token X. Let us say that he wants to swap Δy of the Y token for an amount Δx of the token X from the pool. The number of token X becomes $x - \Delta x$ after this transaction. The swapper must pay a proportional fee $\gamma \Delta y$, which is automatically reinvested in the pool for liquidity provision by the protocol design and is distributed to providers only when they exit the pool.[10] Thus, only an amount $(1 - \gamma)\Delta y$ is used for token swapping. Put another way, the swapper deposits Δy into the pool and then the protocol divides this into two parts, the amount $(1 - \gamma)\Delta y$ used for swapping—according to the bonding curve—and the transaction fee $\gamma \Delta y$, which is automatically reinvested into the pool only *after* the transaction.[11] Therefore, according to the bonding curve (1), we must have

$$(x - \Delta x)(y + (1 - \gamma)\Delta y) = L^2. \tag{2}$$

Since the trading fee $\gamma \Delta y$ is automatically reinvested in the pool, the total reserves of token Y after swapping is $y + \Delta y$. This way, the price changes from P to P', where:

$$P' = \frac{y + \Delta y}{x - \Delta x}.$$

We now introduce a novel way to measure liquidity for AMMs by applying the traditional market liquidity measure of Kyle (1985) to this analysis. That is, we take the price change from P to P' and divide this by the trade volume as measured by Δy, obtaining:

$$\lambda = \frac{P' - P}{\Delta y} = \frac{1 + (1 - \gamma)(1 + \Delta y / y)}{x}. \tag{3}$$

Note that λ is essentially an *ill*iquidity measure because the larger λ is, the less liquid the market is. Now, by first rewriting Equation (2) and then using Equation (3), we can express the quantity Δx of token X received by the swapper as:

$$\begin{aligned} \Delta x = x - \frac{L^2}{y + (1 - \gamma)\Delta y} &= \frac{\Delta y}{P} \cdot \frac{1 - \gamma}{1 + (1 - \gamma)\Delta y / y} \\ &= \frac{\Delta y}{P} \cdot \frac{1 - \gamma}{\lambda x - (1 - \gamma)}. \end{aligned} \tag{4}$$

This way, we see that Δx consists of two components, namely (i) the amount of token X bought at the current price, that is, $\Delta y / P$, and (ii) the pool *price slippage* $(1 - \gamma)/(\lambda x - (1 - \gamma))$. Note that the price slippage term decreases with both the proportional fee γ

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and λ; it is bounded above by 1 when $\gamma = 0$ and $\Delta y = 0$, and it decreases to zero as $\gamma \to 1$. Therefore, the price will change after any token swap with $\Delta y > 0$. Because of this positive price slippage term, the combined value of the deposited tokens in the pool will always be less than their combined value if the tokens were never deposited in the first place. The difference in the combined token value is called the *impermanent loss* to providers, and it would be realized if they were to withdraw their tokens at any price different from the price at which they deposited the tokens. Therefore, to encourage liquidity provision, the income from trading fees needs to exceed this impermanent loss.

Providers in the Uniswap markets have suffered severe impermanent loss over time. However, v3 does nothing to alleviate this problem. In fact, Aigner and Dhaliwal (2021) show that impermanent loss is always greater in v3 than it is in v2. Empirical verification of the excess impermanent loss from v3, relative to v2, is also provided by Loesch et al. (2021), who estimate an impermanent loss of about 260 million USD during the first few months of operation of v3, between May and September 2021. During that time, almost 50% of providers experienced negative total returns. To hedge these losses, Clark (2020), Fukasawa et al. (2023), and Deng et al. (2023) derive static replication strategies using ether options.

Because both tokens should be available to swap at any price P, the issue of over-capitalization with v2 has long been recognized; indeed, inefficient use of capital was the main motivation for adding the new v3 protocol to the set of Uniswap AMMs. Milionis et al. (2022) and others[12] argue that v3 has better capital efficiency because in v2 only small fractions of the total reserves of each token, namely $\Delta x/x$ and $\Delta y/y$, are utilized for swapping Δx of token X forΔy of token Y. However, improved capital efficiency does not imply a higher level of market efficiency, in terms of incorporating new information from within and outside the market. So it is natural to ask: Do the technological advances made in v3 actually improve market efficiency, relative to v2?

The main technological improvements of the v3 protocol address the issue of capital efficiency by allowing liquidity providers to specify a finite price interval over which they are willing to supply the swappers. The protocol still uses a CPMM, but now providers have a greater selection of pools according to the fee tiers set by the protocol. Whereas v2 only has one pool for each token pair, there are several pools in v3 offering different fee tiers. For instance, at the time of writing, there are four pools for USDC-ETH, with fees $\gamma = 0.01\%$, 0.05%, 0.3% and 1%. Some pairs have even more pools. Having selected a pool, providers then set a price range, say $[P_l, P_u]$, within which they are willing to provide liquidity.

The ability to select both liquidity ranges and fees in v3 offers more control over the trade-off between trading volume and impermanent loss, since both decrease as the fee increases. But does this much greater flexibility for providers also improve market efficiency? Because v3 has more pools per token pair than v2 and trading is more dispersed on v3, ceteris paribus, this should decrease the market efficiency for v3 relative to v2. However, the improved capital efficiency of v3 attracts more providers overall. To see this for the USDC-ETH token pair, Figure 2 compares the trading volume and illiquidity on Uniswap v2 with that on the 0.05% and 0.3% pools for v3. On the left, Figure 2a shows that the trading volume on the v3 pools is roughly five times that of v2, with almost all trading occurring on the lower-fee pool. On the right, Figure 2b depicts the illiquidity defined as the price change divided by the trading volume.[13] Figures are calculated daily and the illiquidity time series are also smoothed for better visualization. The mean illiquidity of v3 over the sample period is 0.005, compared with a mean of 0.030 for v2, which is six times larger. The increase in trading volume and liquidity indicates directly that the v3 market is deeper and more active than v2.

Because liquidity providers receive liquidity tokens that can be used elsewhere in decentralized finance, it is common to use the terms *minting* and *burning* liquidity for the provision and removal of liquidity.[14] Once liquidity is provided over a given price range, the provider cannot change this range without incurring gas fees—these are required for both providing and removing liquidity via the smart contract. For this reason, providers try to avoid frequent changes to their liquidity range. But when prices are highly volatile, the decision about liquidity provision becomes rather complex (Chen et al. 2024). As long as the price remains within the range

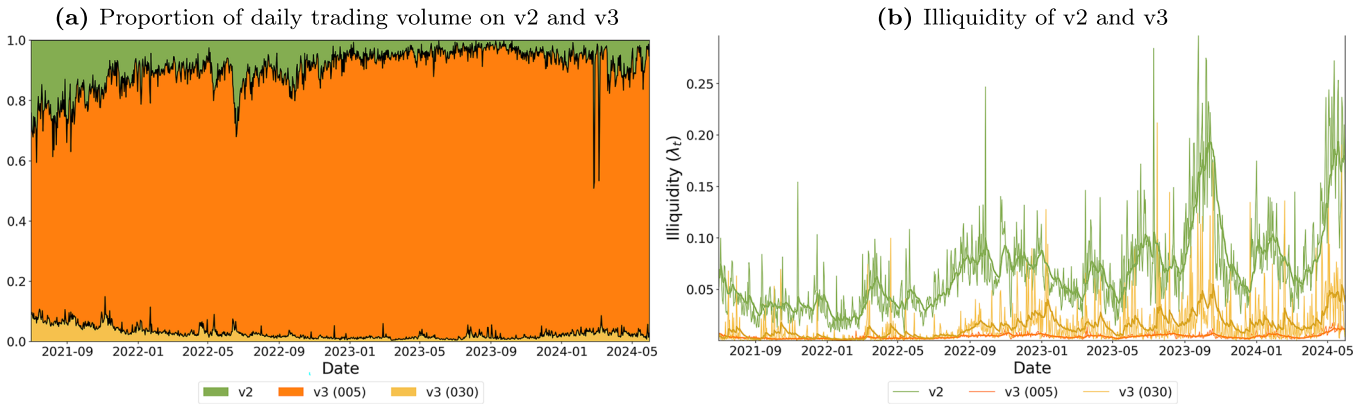

FIGURE 2 | Daily trading volume and illiquidity of v2 and v3, on the USDC-ETH with 0.05% and 0.3% fee rates. Our sampling period is between July 1, 2021 and June 1, 2024. Similar to Kyle (1985), we measure illiquidity by the ratio of price change to trading volume, that is, $\lambda_t = (\text{Price}_t - \text{Price}_{t-1})/\text{Trading Volume}_t$. To present the trend more clearly, we also display series (in bold lines) that are smoothed by an exponential moving average: $\nu_t = 0.1a_t + 0.9\nu_{t-1}$, where a_t (ν_t) is the original (smoothed) sequence of illiquidity.

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$[P_l, P_u]$, the provider will earn transaction fees from swappers, but if the price moves out of this range, the provider's liquidity position is deactivated.[15] As argued by Adams et al. (2021) and Loesch et al. (2021), liquidity provision via range orders in v3 resembles the strategic market making of a limit order book: buying token X moves the price up toward the boundary P_u, selling token X moves the price down toward P_l, if the price moves out of the range the liquidity position becomes dormant, and if the price returns to the provider's range it is automatically reactivated. Our data also shows that providers on v3 supply more liquidity and adjust and monitor their positions far more frequently, indicating a much deeper market than v2. For all these reasons, v3 may attract different types of traders, specifically market makers in large hedge funds and institutions who have better active liquidity management skills;[16] and the presence of more informed traders on v3 might improve its efficiency.

4 | Data and Statistical Tests

According to the criteria used by CoinMarketCap, after the collapse of FTX, Coinbase, Binance, and Bitstamp became the largest CEXs for trading ether spot markets. Ether is important because it has become the numéraire for decentralized finance, being the unit for gas fees of smart contracts on the dominant (Ethereum) chain, and the base currency for non-fungible token transactions. Bitcoin is a main focus for manipulative trades on the many unregulated CEXs, and some huge "whale" trades on certain products like the Binance perpetuals completely dominate the bitcoin price discovery process (Peterson 2021). So, rather than focus on bitcoin, we analyze the relative price discovery and efficiency of ether (ETH) traded on the major CEXs and DEXs, selecting Coinbase and Bitstamp to represent the centralized exchanges. Because these are among the few CEXs that are semi-regulated, in that they have licenses to operate in all major US states and Europe. We have excluded Binance due to significant gaps in their data over extended periods, as detailed in Appendix A.

Our research focuses on 11 token pairs. The first is USDC-ETH on Uniswap markets versus USD-ETH on Coinbase and Bitstamp. Then we identified 10 matched pairs that are actively traded across Uniswap v3 pools 001, 005, and 030 (which have fee tiers 0.01%, 0.05%, and 0.30%, respectively), Uniswap v2, Bitstamp, and Coinbase. Specifically:[17]

- ETH-USDC, ETH-USDT, BTC-ETH, ETH-DAI, and ETH-LINK are the only pairs traded across Uniswap v2, Uniswap v3, Coinbase, and Bitstamp.
- BTC-USDC, DAI-USDC, LINK-USDT, BTC-USDT, and BTC-UNI are traded exclusively on Uniswap v3, Coinbase, and Bitstamp.

Our sample period is from July 1, 2021 to June 1, 2024. We start the sample on July 1 to eliminate the interference of market instability immediately following Uniswap v3's release on 5 May 2021. Trade-level data on Uniswap markets are obtained from CCdata. These data contain a complete record of each mint, burn, and swap transaction, including the timestamp and the amount of each token swapped in or out. We then aggregate the trade-level data to minute-level by taking the first trade price and the total trading volume within each minute. We downloaded minute-level data directly from the Coinbase and Bitstamp official websites. We considered using higher frequency data, such as at 15-s intervals, but the data were too sparse even for the most liquid token pair, ETH-USDC. Furthermore, much existing price-discovery literature supports the use of minute-level, or even coarser data intervals.[18]

Table 1 reports the average daily trading volumes on different platforms, namely those where there are order books and/or pools available for the token pair. Coinbase, as the number one exchange for spot ether trades, has a much larger trading volume than the other platforms. Bitstamp's trading volume is comparable to that of Uniswap v2 and Uniswap v3 (030). But for most other pairs, Uniswap exhibits the greatest trading volumes. Table C1 in Appendix C summarizes some additional preliminary statistics of the data on all token pairs. Table C1a presents the results of the Dickey and Fuller (1979) test on minute-level log price and return series, which show that all log return series are stationary. Then Table C1b reports the results of Johansen (1988) tests, so that we make a reasonable assumption about the cointegration rank of the log prices, which is required for the estimation of a suitable vector error correction model (VECM) in Section B. Almost all token pairs have cointegration rank $k-1$, with k denoting the number of exchanges the

TABLE 1 | Comparison of trading volumes.

	v3 (001)	v3 (005)	v3 (030)	v2	Bitstamp	Coinbase
ETH-USDC vs ETH-USD		216.93	28.68	8.07	33.33	406.69
ETH-USDC		216.93	28.68	8.07	0.23	
ETH-USDT		43.32	13.12	6.92	0.64	22.42
BTC-ETH		31.23	11.36	0.45	2.12	17.40
BTC-USDT			1.28		1.23	28.53
BTC-USDC			5.73		0.72	
DAI-USDC	19.92	4.55				
ETH-DAI		13.64	5.64	1.24		1.18
ETH-LINK			4	0.4		0.48
UNI-BTC			0.01		0.03	0.65
LINK-USDT			0.01			0.22

Note: Average daily trading volume (in million USD) on Uniswap v2, Uniswap v3, Bitstamp, and Coinbase, between July 1, 2021 and June 1, 2024.

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pair is traded on. To align with our VECM, we conducted additional tests on the intra-day minute-level data for each day and calculated the proportion of null hypotheses rejected for different cointegration ranks. The findings remain consistent over the entire sample, underscoring the robustness of our results.

5 | Empirical Results

Table 2 summarizes the empirical tests we adopt to investigate the price discovery ability and level of market efficiency of the exchanges. The tests are explained in detail in Appendix B. We apply them to minute-level spot returns on each day, repeating them daily across the entire sample and summarize our results in Tables 3–5, and Figure 3. Table 3 reports the two-way Granger causality test results, based on bivariate VECMs that are calibrated separately on each day of the sample; Table 4a reports the median CS (component share) and MIS (modified information share) derived from the same models; Table 4b reports the net spillover from each of the four exchanges, the gross spillover of exchanges are reported in Table C2 in Appendix C; Table 5 reports the results of the Ljung-Box test, AVR test, and Bartels' test; and Figure 3 depicts time series of the R/S Hurst exponents for the main token pairs, with exponents for other token pairs depicted in Figure C2 in Appendix C.

Several conclusions may be drawn from these results. First, from Table 3 we see that Coinbase and Bitstamp are generally more efficient than Uniswap. The following observations justify this statement: (i) Granger causality test results indicate that both these exchanges frequently lead prices on Uniswap v2 and Uniswap v3. For the ETH-stablecoin token pairs, there is also strong evidence for bidirectional causality from Uniswap v3 (005) to Coinbase and Bitstamp, but Uniswap v2 plays much less of a leadership role; (ii) Table 4a shows that Coinbase's median CS and MIS are the highest, for all token pairs that are traded there, and for the other pairs Uniswap v3 (005) has greater median CS and MIS than Bitstamp. However, Uniswap v3 (030) has the smallest median CS and MIS. These findings are relatively stable over the sample—see, for example, Figure C1, which depicts the evolution of daily CS and MIS values of exchanges for ETH-USDC versus ETH-USD; (iii) Regarding the median net spillover rates reported in Table 4b, Bitstamp and Coinbase are almost always the main "senders" of information shocks. Figure 4 illustrates an example comparing ETH-USDC and ETH-USD, where both CEXs show positive net spillovers on each day of our sample, whereas the Uniswap markets very often exhibit negative net spillovers; (iv) Also, for seven of the eleven pairs the Uniswap pools fail our statistical tests on market efficiency (namely the Ljung-Box, AVR and Bartels' tests) more frequently than Coinbase and Bitstamp do, as reported in Table 5; and furthermore (v) Figure 3 shows that the R/S Hurst exponents of token pairs on both Coinbase and Bitstamp tend to be closer to 0.5 than the equivalent exponents for Uniswap.

Secondly, comparing price discovery ability only between the different Uniswap pools, we find that Uniswap v3 (005) prices tend to lead those on Uniswap v2 and v3 (030). To see this, note that: (i) the CS and MIS of Uniswap v3 (001 and 005) in Table 4a are almost always higher than the CS and MIS for other pools; (ii) from Table 4b, Uniswap v3 (001 and 005) often generates positive spillover to other exchanges, and passes the Ljung-Box, AVR and Bartels' test more often in Table 5; and (iii) for three out of 10 pairs traded on v3 (030), the daily R/S Hurst exponents shown in Figure 3 can go beyond 0.65 (indicating a state persistence) or below 0.45 (indicating a reverse state persistence) for long periods. But the R/S Hurst exponent for token pairs traded on Uniswap v3 001 and v3 005 pools are almost never state persistent, except for DAI-USDC, where the R/S Hurst exponent indicates strong reverse state persistence until the end of 2023—see Figure C2 in Appendix C. Overall, these findings provide strong evidence of a positive effect from the technological advances in Uniswap v3. The improvement in price discovery ability for Uniswap v3 (001 and 005) confirms that more sophisticated and institutional traders are using Uniswap v3, as reported by Coinmarketcap.[19]

Thirdly, price discovery on Uniswap v3 (005) often exceeds that for Bitstamp, even for USD(C)-ETH, where Bitstamp is the third-largest CEX. To support this observation: (i) the Granger causality results in Table 3 show that v3 (005) and Bitstamp lead other exchanges equally often; (ii) from Table 4a the median CS and MIS values of v3 (005) are higher than those of Bitstamp for

TABLE 2 | Summary of statistical tests.

Category	Test	Literature
Price discovery	Two-way Granger (1969) Causality Test	Procasky (2021)
	Gross and Net Spillover	Pesaran and Shin (1998); Alexander et al. (2020, 2022)
Price discovery/Market efficiency	Component share (CS)	Gonzalo and Granger (1995); Hasbrouck (1995); Lien and Shrestha (2009); Alexander et al. (2020, 2022)
	Modified information share (MIS)	Hasbrouck (1995); Lien and Shrestha (2009); Alexander et al. (2020, 2022)
Market efficiency	Ljung and Box (1978) test	Urquhart (2016)
	Wild-bootstrapped Automatic Variance (AVR) test	Andrews (1991); Choi (1999); Charles et al. (2012); Urquhart (2016)
	Bartels (1982)'s rank test	Urquhart (2016)
	Rescaled Hurst (R/S Hurst) exponent	Hurst (1951); Urquhart (2016)

Note: The tests we adopt to analyze the price discovery ability and market efficiency of the four crypto exchanges, Uniswap v2, Uniswap v3, Coinbase, and Bitstamp.

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TABLE 3 | The percentage of days during our sample for which bivariate Granger causality tests are rejected at the 1% significance level.

BTC, ETH vs USD.

	From	v3 (005)	v3 (030)	v2	Bitstamp	Coinbase
	To					
ETH-USDC vs ETH-USD	v3 (005)	—	16.42	31.61	99.72	98.97
	v3 (030)	99.06	—	94.84	98.87	98.87
	v2	99.34	33.58	—	99.25	99.16
	Bitstamp	90.9	7.5	19.7	—	41.65
	Coinbase	96.81	8.35	24.39	99.62	—
ETH-USDC	v3 (005)	—	16.42	31.61	17.92	
	v3 (030)	99.06	—	94.84	68.01	
	v2	99.34	33.58	—	37.34	
	Bitstamp	94.65	46.81	81.71	—	
ETH-USDT	v3 (005)	—	44.65	55.63	79.64	99.91
	v3 (030)	98.59	—	96.62	95.03	99.44
	v2	98.31	27.49	—	85.37	99.62
	Bitstamp	97.65	30.49	58.35	—	99.44
	Coinbase	80.21	6.75	15.85	51.22	—
BTC-ETH	v3 (005)	—	19.00	20.74	99.34	99.34
	v3 (030)	63.24	—	30.84	65.36	66.31
	v2	50.58	43.22	—	53.59	54.84
	Bitstamp	56.47	9.88	10.47	—	99.72
	Coinbase	26.83	7.68	8.33	95.03	—
BTC-USDT	v3 (030)		—		84.88	86.38
	Bitstamp		8.17		—	99.44
	Coinbase		4.98		61.73	—
BTC-USDC	v3 (030)		—		84.05	
	Bitstamp		14.07		—	

Other tokens.

	From	v3 (001)	v3 (005)	v3 (030)	v2	Bitstamp	Coinbase
	To						
DAI-USDC	v3 (001)	—	2.46				
	v3 (005)	3.68	—				
ETH-DAI	v3 (005)		—	53.9	63.23		82.27
	v3 (030)		82.07	—	82.82		85.35
	v2		89.02	33.99	—		87.80
	Coinbase		71.2	16.43	30.68		—
ETH-LINK	v3 (030)			—	26.92	95.31	
	v2			79.08	—	86.21	
	Coinbase			32.08	18.86	—	
UNI-BTC	v3 (030)			—		24.51	40.47
	Bitstamp			14.38		—	72.13
	Coinbase			6.98		9.81	—
LINK-USDT	v3 (030)			—			30.99
	Coinbase			5.92			—

Note: The null hypothesis of no causality from one exchange to another is tested daily, and the percentage of days on which the test is rejected is reported. Our sample is from July 1, 2021 to June 1, 2024, in total 1066 trading days.

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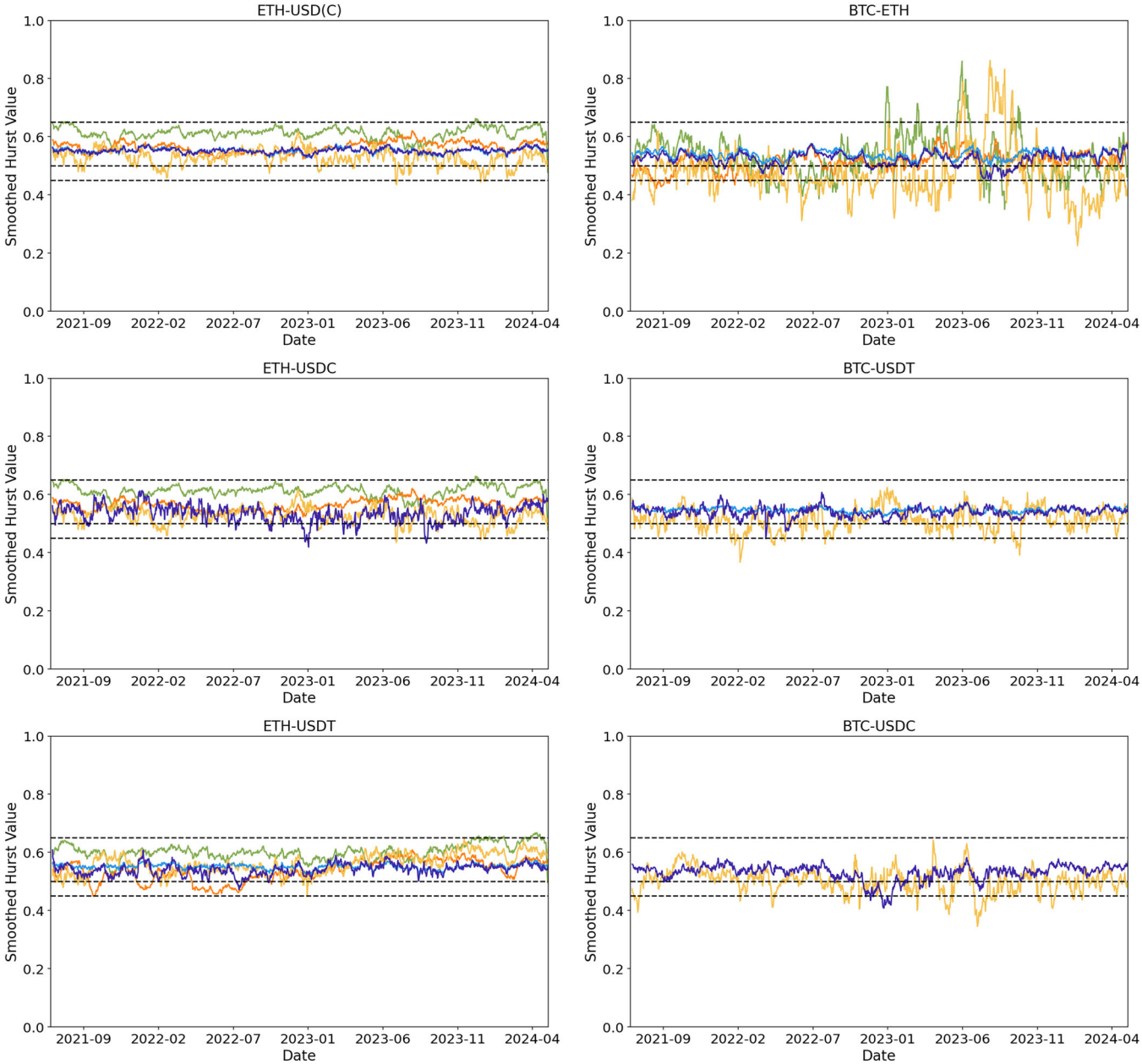

FIGURE 3 | R/S Hurst exponents for major token pairs. The R/S Hurst exponent value staying between 0.65 and 0.45 indicates no state persistence in the timeseries; a value above 0.65 (below 0.45) indicates a (reverse) state persistence. Our sample period is between July 1, 2021 and June 1, 2024, a total of 1066 trading days. To present the trend more clearly in these plots, the series is smoothed by an exponential moving average: $v_t = 0.1a_t + 0.9v_{t-1}$.

three out of four pairs traded on both exchanges; (iii) while Bitstamp seems always to generate positive net spillover in Table 4b, v3 (005) achieves the same for three out of six pairs that are traded there. In fact, for USD(C)-ETH, there are many days when v3 (005) generated positive net spillovers to Bitstamp—see Figure 4; and (iv) Uniswap v3 (005) fails the Ljung-Box, AVR and Bartels' tests less often than Bitstamp, for two out of the four pairs that are traded in both exchanges, as reported in Table 5.

Finally, our comparison of different Uniswap v3 pools shows how the fee structure impacts price discovery ability and efficiency. The 0.01% DAI-USDC pool v3 (001) is designed for pairs having low volatility, and it is marginally more efficient than v3 (005); but the 0.05% fee pools are also more efficient than the 0.3% fee pools and the v2 pools. The v3 (030) pools are more suited to token pairs that are highly volatile, reflecting the inherent trade-off between the fees that providers receive and the more frequent need to pay gas fees to adjust the price range for liquidity supply. These findings highlight the nuanced role of fee selection in shaping the price discovery contribution of Uniswap v3 markets. Our results align with the expectation that low-volatility pairs gravitate toward lower fee tiers (0.01% and 0.05%) and that these tiers tend to provide more efficient trading environments. Conversely, higher volatility pairs gravitate towards the 0.30% tier, where efficiency is moderated by the higher fee structure.

In Section 3, we explained how technological improvements in the v3 protocol addressed the issue of capital efficiency and

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TABLE 4 | Median values of the daily MIS and CS shares and daily net spillover from different exchanges.

(a) Median CS and MIS. BTC, ETH vs USD.

		v3 (001)	v3 (005)	v3 (030)	v2	Bitstamp	Coinbase
ETH-USDC vs ETH-USD	CS		0.227	0.099	0.087	0.084	0.435
	MIS		0.228	0.018	0.025	0.209	0.472
ETH-USDC	CS		0.629	0.095	0.101	0.112	
	MIS		0.843	0.018	0.034	0.048	
ETH-USDT	CS		0.120	0.099	0.101	0.107	0.499
	MIS		0.135	0.019	0.029	0.076	0.681
BTC-ETH	CS		0.055	0.170	0.156	0.122	0.350
	MIS		0.026	0.047	0.057	0.180	0.530
BTC-USDT	CS			0.160		0.181	0.605
	MIS			0.039		0.153	0.749
BTC-USDC	CS			0.222		0.778	
	MIS			0.072		0.928	

Other tokens.

		v3 (001)	v3 (005)	v3 (030)	v2	Bitstamp	Coinbase
DAI-USDC	CS	0.735	0.265				
	MIS	0.753	0.247				
DAI-ETH	CS		0.147	0.110	0.140		0.491
	MIS		0.178	0.021	0.043		0.650
LINK-ETH	CS			0.298	0.106		0.552
	MIS			0.112	0.027		0.827
UNI-BTC	CS			0.255		0.131	0.478
	MIS			0.111		0.057	0.683
LINK-USDT	CS			0.287			0.713
	MIS			0.150			0.850

(b) Median net spillover (in %). BTC, ETH vs USD.

From	v3 (001)	v3 (005)	v3 (030)	v2	Bitstamp	Coinbase
ETH-USDC vs ETH-USD		−2.53	−1.53	−5.46	5.63	4.78
ETH-USDC		0.03	−0.36	−0.13	0.60	
ETH-USDT		1.90	−1.04	−3.57	1.28	1.54
BTC-ETH		−1.01	−0.15	0.01	0.73	0.29
BTC-USDT			−0.03		1.29	−1.19
BTC-USDC			−0.00		0.00	

Other tokens.

From	v3 (001)	v3 (005)	v3 (030)	v2	Bitstamp	Coinbase
DAI-USDC	0.00	−0.00				
ETH-DAI		0.01	−0.04	−0.08		0.03
ETH-LINK			0.03	0.01		−0.06

(Continues)

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TABLE 4 | (Continued)

Other tokens.						
From	**v3 (001)**	**v3 (005)**	**v3 (030)**	**v2**	**Bitstamp**	**Coinbase**
UNI-BTC			0.00		0.00	−0.02
LINK-USDT			−0.00			0.00

Note: Our sample is from July 1, 2021 to June 1, 2024, in total 1066 trading days. The daily MIS and CS represent price discovery shares, derived from our daily VECM model calibrations. Panel B tabulates the long-horizon (h = 60 min) median net spillover from different exchanges, computed by first measuring the gross spillover of market X to all other markets on a given day, and subtracting from this the gross spillover that market X receives from all other markets on the same day, and then reporting the median of these net daily spillovers. Color indicators are included in the results. For panel (a), the dark red (green) lights indicate the highest (lowest) values, and the light red (green) lights indicate the second highest (lowest) values. For panel (b), the dark red lights indicate positive net spillover values.

TABLE 5 | Results of Ljung-Box, AVR, and Bartels' tests.

BTC, ETH vs USD.						
	v3 (001)	**v3 (005)**	**v3 (030)**	**v2**	**Bitstamp**	**Coinbase**
ETH-USDC vs ETH-USD						
Ljung-Box		60.04	33.02	61.42	20.54	21.95
AVR		43.81	40.43	60.57	11.35	15.20
Bartels'		87.8	99.91	95.68	8.91	15.48
ETH-USDC						
Ljung-Box		60.04	33.02	61.42	76.55	
AVR		42.87	40.06	60.57	75.33	
Bartels'		87.80	99.91	95.68	81.52	
ETH-USDT						
Ljung-Box		69.42	32.83	60.40	31.80	18.39
AVR		46.53	41.28	55.60	27.86	12.29
Bartels'		65.57	98.50	91.18	33.02	13.51
BTC-ETH						
Ljung-Box		27.49	83.30	98.50	37.62	30.39
AVR		21.20	84.33	98.31	38.93	23.26
Bartels'		69.61	0100	98.59	34.24	26.64
BTC-USDT						
Ljung-Box			93.15		39.77	18.11
AVR			92.59		34.15	11.54
Bartels'			98.03		42.5	11.35
BTC-USDC						
Ljung-Box			78.80		41.56	
AVR			79.64		34.90	
Bartels'			97.84		45.31	
Other tokens.						
	v3 (001)	**v3 (005)**	**v3 (030)**	**v2**	**Bitstamp**	**Coinbase**
DAI-USDC						
Ljung-Box	58.82	98.92				
AVR	67.74	99.25				
Bartels'	45.38	99.25				
ETH-DAI						
Ljung-Box		39.49	68.11	27.34		60.98
AVR		26.55	69.98	27.63		56.66

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TABLE 5 | (Continued)

Other tokens.	v3 (001)	v3 (005)	v3 (030)	v2	Bitstamp	Coinbase
Bartels'		32.36	95.03	37.43		62.01
LINK-ETH						
Ljung-Box			47.19	80.86	57.41	
AVR			54.41	79.34	54.88	
Bartels'			66.51	79.17	77.02	
UNI-BTC						
Ljung-Box			0100		86.31	34.94
AVR			0100		86.50	33.81
Bartels'			0100		86.12	45.33
LINK-USDT						
Ljung-Box			0100			40.86
AVR			0100			44.18
Bartels'			0100			58.31

Note: We report the percentage of days in our sample for which at least one of the Ljung-Box, AVR, or Bartels' tests are rejected at the 1% significance level. Our sample is from July 1, 2021 to June 1, 2024, in total 1066 trading days. Color indicators are included in the results. The dark red (green) lights indicate highest (lowest) values, and the light red (green) lights indicate second highest (lowest) values.

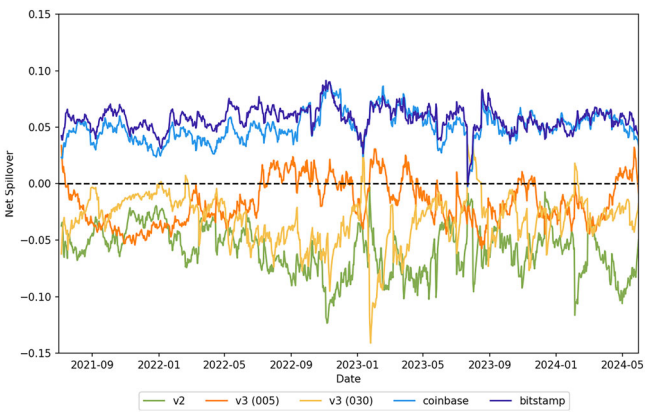

FIGURE 4 | Daily net spillover. Daily series of net spillovers (in %) for ETH-USDC on Uniswap versus ETH-USD on Coinbase and Bitstamp, computed using daily-calibrated VECMs. Our sample is from July 1, 2021 to June 1, 2024, in total 1066 trading days. To present the trend more clearly, the series is smoothed by an exponential moving average for plotting: $v_t = 0.1a_t + 0.9v_{t-1}$.

questioned whether this would lead to increased trading volumes and improved liquidity in v3 pools. In the following we answer this question empirically, by estimating multi-variate linear regression models:

$$y_{it} = \alpha_i + \sum_{j=1}^{5} \beta_{ij} x_{ijt} + \epsilon_{it}, \tag{5}$$

where y_{it} is either (a) the price discovery ability of v2 ($i = 1$) and v3 ($i = 2$) markets, represented by the CS and MIS measures computed in Table 4a, or (b) the market efficiency of the pool represented by the absolute deviation of the R/S Hurst exponent from 0.5 (variable ADH), whose daily time series are depicted in Figure 3. Note that the CS and MIS represent the price discovery contribution of pool i to the entire system of pools and exchanges where the same token pair is traded, whereas the efficiency variable ADH just depends on the individual pool.

The explanatory variables x_{ijt} for each token pair modeled using Equation (5) are: Vol ($j = 1$), the volatility of the token pair minute-level log returns in pool i on day t; Basis ($j = 2$), defined as the absolute value of the average price difference between pool i prices and the CEX market prices, averaged over every minute-level observation on each day t; LTR ($j = 3$), the large trade ratio, computed as the ratio of the sum of the volumes of large trades (above the 75% quantile size per day) to the total trading volume in pool i on day t; the illiquidity measure λ defined in Equation (3) ($j = 4$), which is computed as explained in Figure 2; and LPA ($j = 5$), the activity of liquidity providers, which is proxied by the daily average minute-level change in the liquidity supply constant L, for pool i on day t.

Table C3a in Appendix C examines the correlations between the five explanatory variables and trading volume. We exclude the daily trading volume because it is very highly correlated with both volatility and the large trade ratio in every pool. The five explanatory variables we have selected have reasonably low correlations in most pools. Results using trading volume instead of volatility are qualitatively similar, and are available in Table C4 in Appendix C. The only issue with multi-collinearity might be between basis and volatility (see Table C3b); for example, in the LINK-USDT pool on v3 (030), these have a correlation of 0.597.

The impact that some of these explanatory variables have on liquidity supply and trading on decentralized crypto exchanges has already been studied, to some extent. For instance, while Capponi and Jia (2021), Chen et al. (2024), and Lehar and Parlour (2024) all show that volatility affects both liquidity provision and impermanent loss, the net effect

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on price discovery and efficiency remains unclear. Uniswap offers several pools with different fee levels, and it may be that lower-fee pools have relatively lower liquidity during volatile periods compared with higher-fee pools. This is because supply ranges need to be changed more frequently when markets become more volatile, and changing the supply range incurs gas costs, so to compensate for these increased costs, liquidity providers could shift to higher-fee pools at such times. And if pools become less efficient or less dominant in price discovery during volatile periods, this implies that *informed* traders move away from Uniswap and toward centralized exchanges at such times. Nevertheless, because the trading volume-volatility correlation is very high in every pool, as shown in Table C3a in Appendix C, *uninformed* speculative trading volume increases on Uniswap during volatile periods.

Regarding the basis and the prevalence of large trades as explanatory variables, Heimbach et al. (2022a) show that Uniswap prices may deviate further from centralized exchanges during periods of extreme distress. Hence, traders attempting to exploit arbitrage opportunities between centralized exchanges and Uniswap pools should become more active during periods when the basis is large. If these arbitrageurs are informed professional traders (who tend to make larger trades) a large basis could improve the price discovery ability of the Uniswap pool; but if they are speculators (who tend to make smaller trades) the opposite could be the case.

Lastly, as well as investigating the explanatory role of all these variables for price discovery and efficiency, we ask whether their effects differ according to the fee tier of the liquidity pool—a question that is in line with previous research in traditional finance markets, which finds that price discovery ability varies under different market conditions (Chakravarty et al. 2004; Entrop et al. 2020).

Table 6 reports our main results. First, note that during volatile periods, price discovery ability (CS, MIS) and efficiency (ADH) of almost all the Uniswap pools reduce. This implies that informed traders prefer to use centralized exchanges when markets become volatile, but, because total trading volume invariably increases when markets are volatile, it must be that more speculators start trading in the Uniswap pools at these times. The only exception is the DAI-USDC v3 (001) pool, which is not traded on our centralized exchanges (hence there is no basis variable). However, this stablecoin pair has very low volatility, and our results merely show that the low-fee pool becomes more dominant and efficient compared with the higher-fee pool v3 (005). The basis between Uniswap and Coinbase or Bitstamp also has a predominately negative effect on price discovery. Since the basis tends to widen during volatile periods, this supports our finding that fewer informed traders, but a greater number of uninformed traders, use Uniswap when markets become volatile. The basis has a mixed and less significant effect on efficiency; the only notable result being the coefficient of 2.537 for the UNI-BTC v3 (030) pool, but this pool has a very low trading volume (of about 10,000 USD per day, on average during our sample).

The effect of large trades, as captured by the LTR variable, depends on the fee structure. For instance, large trades decrease the price discovery ability of the v3 (005) pools but increase the price discovery of the larger v3 (030) and v2 pools. If large trades bring more information to the market and thereby improve market efficiency and price discovery, this shows that more informed traders choose to trade first on higher-fee pools where there is sufficient liquidity to do the trade without too much slippage. Regarding the relative efficiency, when markets are highly volatile, the v3 liquidity provider needs to either (a) widen their supply range, in which case the capital efficiency decreases and the pool becomes more like the v2 pools, or (b) change a narrow supply range more frequently, thus incurring gas fees. Either action will decrease market efficiency. Since the higher-fee v3 (030) pools are designed to compensate liquidity providers for the more frequent supply range changes, which volatile token pair prices require, providers may switch supply to these higher-fee pools during especially volatile periods, thus increasing liquidity and attracting greater trading volume. However, the significant *decrease* in CS, MIS, and ADH during highly uncertain periods indicates that more speculators, rather than informed swappers, may be using the higher-fee pools at such times, when there are also more opportunities for cross-exchange arbitrage.

The illiquidity variable typically has a mixed effect: a higher liquidity (when λ is lower) is associated with better price discovery ability but less efficiency in the ETH-USDC v3 (005) pool. But in other pools, the effect is reversed, or not significant. And liquidity provider activity (LPA) also has little significance except in the USDT-ETH v3 (005) pool and the stablecoin pair v1 (001) pool, where, as expected, it has a positive effect on both price discovery and efficiency.

6 | Conclusions

Market efficiency and price discovery ability measure whether the asset prices in a financial market can fully and rapidly incorporate new information from within and outside of the market, respectively. They have significant implications for risk management and hedging, as well as the potential to exploit cross-market arbitrage opportunities. This paper examines the price discovery ability and market efficiency of Uniswap v2 and Uniswap v3 compared with two major centralized exchanges for spot trading, Coinbase, and Bitstamp. We found a total of eleven token pairs that are actively traded in several Uniswap pools as well as on either Coinbase or Bitstamp, or both, and our empirical results are based on almost 3 years of trade-by-trade data from all the platforms listing these token pairs that provide the relevant data. These data are then aggregated to minute-level, from which vector error correction models are estimated day-by-day to generate time series of price discovery and market efficiency metrics for each token pair pool or exchange.

Uniswap markets have attracted much attention from academics and practitioners. The liquidity pools use a basic constant proportion market-making automation in Uniswap v2, and Uniswap v3 employs a more capital and computationally efficient form of automated market maker, which improves liquidity provision. Although both v2 and v3 are less efficient than Coinbase and Bitstamp, the price discovery ability of some v3 pools is now surpassing that of Bitstamp.

The superior price discovery ability of v3, compared with the same token pair pool on v2, can be attributed to the user-defined features of v3. By allowing providers to supply pools

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TABLE 6 | The impact of market characteristics on price discovery and market efficiency.

Coin pair	CS					MIS					ADH				
	Vol	Basis	LTR	λ	LPA	Vol	Basis	LTR	λ	LPA	Vol	Basis	LTR	λ	LPA
v3 (001)															
DAI-USDC	0.009		0.162***	0.009	0.069**	0.032		0.153***	−0.001	0.072**	0.144***		−0.078**	0.161***	0.026
v3 (005)															
USDC/ USD-ETH	−0.001	0.016	−0.244***	0.175***	−0.033	0.117***	0.042	−0.180***	0.279***	−0.027	−0.244***	0.064**	−0.126***	−0.105***	−0.034
ETH-USDC	−0.341***	0.113***	−0.232***	0.124***	−0.033	−0.396***	0.113***	−0.234***	0.123***	−0.032	−0.206***	−0.073**	−0.149***	−0.131***	−0.037
USDT-ETH	−0.340***	−0.075**	−0.124***	−0.037	0.159***	−0.330***	−0.137***	−0.167***	−0.018	0.168***	−0.049	−0.060*	0.005	−0.100***	0.188***
BTC-ETH	−0.055	0.154***	−0.03	−0.174***	0.033	0.056	0.182***	−0.002	−0.157***	0.130***	−0.048	0.032	0.024	0.067*	−0.036
ETH-DAI	−0.340***	−0.011	−0.374***	−0.017	0.051**	−0.339***	−0.018	−0.279***	−0.019	−0.023	0.044	0.024	0.017	0.010	0.055*
DAI-USDC	−0.638***		−0.070**	−0.142***	0.032	−0.592***		−0.060**	−0.116***	0.017	0.289***		−0.002	0.121***	0.024
v3 (030)															
USDC/ USD-ETH	−0.282***	−0.063*	0.111***	−0.060**	0.007	−0.138***	−0.049	0.101***	−0.040	−0.008	−0.166***	−0.031	−0.018	0.018	0.013
ETH-USDC	−0.182***	−0.101***	0.088***	−0.039	0.011	−0.065**	−0.115***	0.077**	−0.025	−0.008	−0.157***	−0.072**	−0.035	0.014	0.012
USDT-ETH	−0.247***	−0.107***	0.044	−0.032	0.023	−0.133***	−0.130***	0.013	−0.020	0.050*	−0.157***	−0.024	0.041	−0.060**	−0.03
BTC-ETH	−0.499***	0.191***	−0.018	−0.001	−0.014	−0.338***	0.357***	−0.015	0.047	−0.034	−0.240***	0.092*	−0.076**	−0.142***	−0.021
BTC-USDT	−0.332***	−0.113***	0.207***	−0.031	−0.033	−0.219***	−0.084***	0.255***	−0.019	−0.039	−0.131***	0.003	−0.031	−0.011	−0.036
BTC-USDC	−0.377***	−0.235***	0.078***	0.023	−0.009	−0.256***	−0.216***	0.087***	0.010	−0.015	−0.174***	−0.019	0	−0.048	−0.011
ETH-DAI	−0.163***	−0.098***	0.117***	0.091***	−0.011	−0.093***	−0.124***	0.098***	0.181***	0.011	−0.099***	−0.047	−0.032	−0.034	−0.001
LINK-ETH	−0.216***	−0.210***	0.142***	−0.143***	0.049*	−0.065**	−0.205***	0.188***	−0.132***	0.051*	−0.140***	−0.066*	−0.068**	−0.017	0.022
LINK-USDT	−0.415***	0.244***	−0.173***	−0.056	0	−0.259***	0.192***	−0.168***	−0.029	0.007	−0.133***	0.182***	−0.031	0.038	−0.019
UNI-BTC	−0.221***	0.310	−0.119***	0.041	−0.018	−0.094***	0.819**	−0.140***	0.104**	−0.019	−0.046	2.537**	−0.06	−0.026	−0.002
v2															
USDC/ USD-ETH	−0.081**	−0.089***	0.084**	−0.038	−0.016	0.085**	−0.090***	0.171***	−0.018	−0.018	−0.038	0.058*	0.093***	−0.047	−0.011
ETH-USDC	0.009	−0.002	0.221***	−0.034	−0.012	0.160***	−0.017	0.285***	−0.016	−0.008	−0.043	0.067**	0.088***	−0.047	−0.01
USDT-ETH	−0.110**	−0.090***	0.108***	0.021	0.005	0.150***	−0.102***	0.127***	−0.066*	0.039	−0.271***	0.091***	0.188***	0.266***	0.025
BTC-ETH	−0.110***	−0.046	−0.150***	−0.057	0.025	−0.084***	0.001	−0.057*	0.106***	−0.005	0.054*	−0.043	−0.055	−0.074	−0.003
ETH-DAI	−0.207***	−0.164***	−0.055*	0.107***	−0.018	−0.120***	−0.151***	−0.021	0.069*	−0.016	−0.092**	0.032	0.057*	0.071*	−0.008
LINK-ETH	−0.238***	−0.076	0.047	0.051	−0.017	−0.053	0.050	0.046	0.006	−0.026	0.019	−0.040	−0.017	0.056	0.03

Note: The results of model (5), which uses minute-level volatility (Vol), basis, large trade ratio (LTR), the market illiquidity measure λ in Equation (3), and the liquidity provider activity (LPA) to explain price discovery ability (CS or MIS) and efficiency (ADH). The sample from July 1, 2021 to June 1, 2024, 1066 observations. All variables are normalized and asterisks ***, **, and * represent 1%, 5%, and 10% significance levels.

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with different fee levels, and to select tailor-made liquidity ranges within each pool, liquidity provision in v3 becomes similar to the limit order book employed in a traditional market—whereas trades in v2 are more like market orders. This potentially attracts more informed traders to the low-fee v3 pools, which consequently demonstrates an improved level of price discovery. Our results have also highlighted the nuanced role of fee selection in shaping the price discovery contribution of Uniswap v3 markets. Uniswap v3 prices in pools with 0.05% fees tend to lead those on Uniswap v2 and v3 pools with 0.30% fees. Low-volatility pairs gravitate toward lower fee tiers (0.01% and 0.05%), and these tiers tend to provide more efficient trading environments. Conversely, higher volatility pairs gravitate toward the 0.30% tier, where efficiency is moderated by the higher fee structure.

Nevertheless, our battery of empirical tests shows that most Uniswap pools remain in the position of follower rather than leader, with prices largely driven by cross-market DEX-CEX arbitrage activities. This is not surprising given that, even adopting layer-two roll-ups designed to speed up transactions, Ethereum consensus and verification mechanisms still slow down information dissemination, which harms the performance of DEXs as efficient financial markets. Unlike traditional markets and the major crypto centralized exchanges, trading volume is often found to have a negative effect on both price discovery and efficiency. This can be explained by the increased presence of speculative rather than informed professional traders during volatile periods, when trading volume increases, and the opportunities for cross-market arbitrage.

Decentralized trading can resolve many of the security and operational challenges faced by centralized exchanges, and with oracles now adopting crypto prices from Uniswap, these liquidity pools are rapidly becoming one of the main pillars of the DeFi community. Although our findings confirm that liquidity pools can be significantly enhanced through changes in smart contract protocols, many more technical developments are needed if they are to attract higher trading volumes.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors obtained data from CCdata, Bitstamp, and Coinbase.

Endnotes

[1] On most CEXs, the same company will act as a custodian, broker, market maker, and central clearing counterparty, not only as the operator of electronic trading platforms that adopt efficient order matching engines and funding management systems to ensure cheap, smooth, and fast transactions. Consequently, they have become the dominant platforms for crypto trading. As of December 19, 2024, the 24-h trading volume of all tokens held on 218 centralized exchanges was 300 billion USD. Source: CoinGecko.

[2] For instance, there are no public records of fiat currency deposits and withdrawals. Only the pseudonymous deposits and withdrawals of crypto are recorded on a blockchain, via connection to the CEXs hot wallet. Also, most CEXs do not provide full details of the limit order book.

[3] For instance, during the hours following its filing for bankruptcy on November 11, 2022, the FTX exchange experienced a possible hack in which hundreds of millions worth of tokens were stolen. Similarly, in October 2022, hackers exploited vulnerabilities to attack Binance, which is by now the largest CEX, and stole BNB tokens worth 570 million US dollars. In 2022, 3.7 billion USD was lost due to various attacks, hacks, and scams, according to CoinDesk.

[4] A gas fee is the amount of native token required for a blockchain network user to conduct a transaction on the network. Nevertheless, see this Solidus Report.

[5] However, a new form of manipulation called maximum extractable value does occur, via sandwich attacks and other forms of front-running on DEXs (Daian et al. 2019; Alipanahloo et al. 2024; Gramlich et al. 2024).

[6] Source: DeFiLlama. The TVL on a DEX depends on the number of tokens in the pools and their USD prices and as such it tracks the market capitalization of crypto, which reached a high of almost four trillion USD at the end of 2024.

[7] Uniswap v4 is due to be launched soon. See the white paper Adams et al. (2023).

[8] An excellent review by Mohan (2022) clarifies the similarities and differences between various types of AMMs, including Constant Mean Market Makers, Constant Sum Market Makers and Hybrid Function Market Makers, as well as CPMMs, which may also be called Constant Function Market Makers, e.g,. as in Milionis et al. (2022).

[9] We only consider pools for swapping two tokens. Multiple token pools do exist, e.g. on Balancer or Curve, but a constant mean market maker (CMMM) is normally used in this case. These pools have reserves $(x_1, x_2, ..., x_n)$ of n tokens $X_1, X_2, ..., X_n$ that satisfy the function $\prod_{i=1}^{n} x_i^{w_i} = K$ with $\sum_{i=1}^{n} w_i = 1$.

[10] Uniswap allows different fee tiers in pools, such as $\gamma \in \{0.01\%, 0.05\%, 0.3\%, 1\%\}$ for the USDC-ETH pair.

[11] Another difference in the Uniswap v3 protocol is that the smart contract automatically mints ERC-721 non-fungible liquidity tokens for fees, and providers then have the *option* to manually redeposit the transaction fees back into the pool after the transaction.

[12] The blog by Austin Adams and Gordon Liao claims that v3 increases capital efficiency for providers by up to 4000x, compared with v2, and that fee returns on non-rebalancing v3 positions outperform comparable v2 positions by more than 50%.

[13] In v3, liquidity provisions are more concentrated around the current pool price. For instance, see the liquidity distribution of the WETH-USDC 0.05% pool. Trades seldom cross multiple liquidity provision intervals. We still use the ratio of price change to trading volume to measure market illiquidity λ in the Uniswap v3 market.

[14] In particular, v2 mints ERC-20 fungible tokens and v3 mints ERC-721 non-fungible tokens, i.e. each token is unique. The rationale behind this change is for liquidity providers to boost income by liquidity mining, see the blog of Adams et al. (2021).

[15] At the boundaries of the price range, liquidity reserves are concentrated on the less valuable asset. For example, if $[P_l, P_u] = [1000, 3000]$ in the USDC-ETH pool, once the price hits 1000, the entire provision becomes ETH tokens, no USDC tokens; and if the price falls further, to 900 say, then the providers balance is measured in ETH. Conversely, if the price appreciates to 3000, then the entire balance is automatically held in USDC.

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[16]See Is UNISWAP v3 Sacrificing Retail for Capital Efficiency?: Flipside Crypto. https://coinmarketcap.com/academy/article/is-uniswap-v3-sacrificing-retail-for-capital-efficiency

[17]LINK-USDC is omitted because it is only traded on Uniswap v3 (fee tier 0.30%) and Binance.

[18]For example, Han et al. (2023) studied the reliability of Uniswap prices from an investor's perspective using 5-min order and trade data, including price discovery measures like Component Share (CS). Similarly, Hansen et al. (2024) examined the price discovery capabilities of Binance, Coinbase, and Uniswap v2 using minute-level data. Studies using tick-by-tick data typically focus on investment or trading strategies. For instance, Capponi et al. (2024) adopted tick-by-tick Ethereum mempool data to investigate investor strategies and distinguish transactions from public and private pools. Hansson (2024) used tick-by-tick data to study how manual, algorithmic, and arbitrage trading, alongside liquidity provision, contribute to price discovery on Uniswap. These examples demonstrate that while finer intervals can be insightful in some contexts, they are not universally adopted and are not, in fact suitable for our study. Foley et al. (2023) suggest how to identify imprecise or slow exchanges by examining whether they record trades with millisecond-level precision, and whether internal latency is an issue, whereby the matching engine speed lags behind the timestamp accuracy (e.g., parts of a market order being processed across multiple milliseconds). However, such reliability concerns do not affect our analysis, as our study is based on lower-frequency data. The data set we utilize is not designed to capture the nuances of millisecond-level precision, but it is well-suited for our research objectives.

[19]op. cit. note 16.

[20]Binance Auto-Convert USDC to BUSD.

[21]Binance Stopped BUSD Auto-Conversion.

[22]ETH-USDC, BTC-USDC Back to Binance.

[23]Conventionally, the VECM model is written as:

$$\boldsymbol{r}_t = \boldsymbol{\alpha}\boldsymbol{\beta}'\boldsymbol{p}_{t-1} + \sum_{q=1}^{Q} \boldsymbol{A}_q \boldsymbol{r}_{t-q} + \boldsymbol{\epsilon}_t,$$

where Q represents the lag order of VECM. For simplicity, our model assumes $Q = 1$, which is well-grounded because in practice, the BIC criterion usually selects a lag order of one (Alexander and Heck 2020).

[24]We compute l following Andrews (1991) using the method specified in the R package designed for this test.

[25]In other words, for the kth sub-sample, $y_{kq} = x_{kq} - \left(\sum_{i=1}^{m} x_{ki}/m\right)$, and $z_{kq} = \sum_{i=1}^{q} y_{ki}$, $q = 1, \ldots, m$.

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Appendix A

Binance Data

On August 29, 2022 03:00 (UTC) Binance introduced its own stablecoin, BUSD, with automatic conversion for users' existing balances and new deposits of USDC stablecoins at a 1:1 ratio.[20] But on March 10, 2023 (UTC) it stopped BUSD auto-conversion for new deposits of USDC,[21] and, on March 12, 2023 06:00 (UTC), it added the ETH-USDC and BTC-USDC pairs back.[22] Because of this, we have missing data on ETH-USDC and BTC-USDC between August 29, 2022 and March 12, 2023. Also, during the periods where Binance does have this data, the trading volume was five to 10 times that of Coinbase and a 100 times that of Bitstamp. Therefore, as expected, Binance has the best price discovery for almost all trading pairs, except BTC-USDT, ETH-USDT (Coinbase is the best for these two). If we include Binance in the spillover analysis, for the periods where data are available, Binance has positive net spillover for the top trading pairs, such as ETH-USDC, ETH-USDT, BTC-ETH, and BTC-USDT, but a negative net spillover for smaller trading pairs for which Coinbase has positive net spillover. For market

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efficiency (including R/S Hurst), Binance generally passes the test, and the Binance market can be generally viewed as informationally efficient.

Appendix B

Statistical Models and Tests

First, we introduce the vector error correction model (VECM, see, e.g., Alexander et al. 2020, for detailed discussion) that is used to explore the price discovery relationship between different liquidity pools, and between liquidity pools and centralized exchanges. Each VECM requires the token pair price system to be cointegrated, which is confirmed by the results in Table C1. We calibrate the VECMs on a daily basis, using the minute-level token pair prices and returns obtained from the k different platforms it is traded on:

$$\boldsymbol{r}_t = \boldsymbol{\alpha}\boldsymbol{\beta}'\boldsymbol{p}_{t-1} + \boldsymbol{A}\boldsymbol{r}_{t-1} + \boldsymbol{\epsilon}_t, \tag{B1}$$

where $\boldsymbol{p}_t$ is a $k \times 1$ vector representing the log price at time t of the token pair on k different platforms; $\boldsymbol{r}_t$ denotes their log returns; $\boldsymbol{\alpha}$ is a $k \times k - 1$ error correction coefficient matrix, whose sign and magnitude represent the reaction direction and speed of the token pair prices when deviating from the long-term equilibrium price of each platform; $\boldsymbol{\beta}$ is a $k \times k - 1$ matrix that makes the $\boldsymbol{\beta}'\boldsymbol{p}_{t-1}$ vector stationary and equal to the deviations from long-term equilibrium; $\boldsymbol{A}$ is a $k \times k$ autoregressive coefficient matrix capturing short-term fluctuations of the ether spot prices;[23] and $\boldsymbol{\epsilon}_t$ denotes the residuals at time t.

We understand how the token pair prices on different platforms influence each other by computing the component and information share measurements for each calibrated VECM. It is worth noting that previous research (such as Hasbrouck 1995; Lien and Shrestha 2009; Alexander et al. 2020) on price discovery testing mainly calibrates bivariate VECMs, as we also do for the Granger causality tests. But two-dimensional VECMs provide limited, one-sided information since the price discovery ability of any given token pair not only varies over time, but also depends on which pools and exchanges are in the system, each corresponding to a different platform where the token pair is traded.

These are the tests we perform day by day, on each calibrated bivariate or k-variate VECM:

1. *Two-way* Granger (1969) *causality test*: For this test, we use bivariate VECMs calibrated using minute-level token pair returns from any two of the pools or exchanges that trade the token pair. The null hypothesis is that the past values of the returns from one pool or exchange (the "origin/from") do not help to predict the future values of the returns from another pool or exchange (the "receiver/to"). Rejecting the Granger test on any specific day indicates that the token pair price on one pool or exchange is leading that of another pool or exchange on that day. Conducting this test across all trading days of our sampling period can help us understand how often the price discovery on one pool or exchange leads that of another pool or exchange.

2. *CS, MIS, gross, and net spillover*: Each token pair has its own k-dimensional VECM, which is calibrated daily. Then following Alexander et al. (2020), we calculate the CS and MIS of the exchange s, denoted CS_s and MIS_s, respectively, as:

$$\mathrm{CS}_s = \frac{\alpha_{\perp,s}}{\sum_{q=1}^{k} \alpha_{\perp,q}}, \ \mathrm{MIS}_s = \frac{([\boldsymbol{\psi}\boldsymbol{F}]_s)^2}{\boldsymbol{\psi}\boldsymbol{\Omega}\boldsymbol{\psi}'},$$
$$\text{with} \quad \boldsymbol{F} = \left(\boldsymbol{G}\boldsymbol{\Lambda}^{-\frac{1}{2}}\boldsymbol{G}'\boldsymbol{V}^{-1}\right)^{-1},$$

on each trading day, where: $\boldsymbol{\Omega}$ denotes the covariance matrix of the residual vector $\boldsymbol{\epsilon}$; $\boldsymbol{\Lambda}$ denotes the diagonal matrix of eigenvalues of $\boldsymbol{\Omega}$ with the corresponding eigenvectors denoted as $\boldsymbol{G}$; $\boldsymbol{V}$ represents the diagonal matrix of the residuals' standard deviations; $\boldsymbol{\psi}$ is the common row vector of $\boldsymbol{\Psi} = \boldsymbol{\beta}_\perp\left[\boldsymbol{\alpha}'_\perp(\boldsymbol{I}_k - \boldsymbol{A})\boldsymbol{\beta}_\perp\right]^{-1}\boldsymbol{\alpha}'_\perp$, where $\boldsymbol{\alpha}_\perp$ and $\boldsymbol{\beta}_\perp$ denote the orthogonal complements of $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$, respectively, and $\boldsymbol{I}_k$ is the $k \times k$ identity matrix. The CS_s statistic indicates the relative contribution of the ether spot price obtained from exchange s to the discovery of the long-term equilibrium spot price of ether. From a different angle, CS_s indicates the relative stability of market s. A high CS_s indicates a good price discovery ability of exchange s. The MIS_s statistic indicates the extent to which exchange s's new information contributes to the variation in the common factor. In other words, MIS_s indicates the speed and accuracy of the ether spot price on exchange s reacting to new information. A high MIS means a quick and complete response of the ether spot price of the exchange to new information shocks. Clearly, these metrics reveal different and complementary characteristics and therefore, are often used together (Yan and Zivot 2010).

The gross spillover from the information shock embedded in the token pair price in pool or exchange j to that in pool or exchange i, denoted θ_{ij}, is also calculated on each trading day, as:

$$\theta_{ij} = \frac{\omega_{jj}^{-1}\sum_{q=0}^{h-1}\left(\boldsymbol{e}_i'\boldsymbol{\Psi}_q\boldsymbol{\Omega}\boldsymbol{e}_j\right)^2}{\sum_{q=0}^{h-1}\boldsymbol{e}_i'\boldsymbol{\Psi}_q\boldsymbol{\Omega}\boldsymbol{\Psi}_q'\boldsymbol{e}_j},$$

where we set h to 60 (minutes looking-ahead), ω_{jj} is the jth diagonal entry of $\boldsymbol{\Omega}$, and $\boldsymbol{e}_j$ is a column vector, which takes the value one in the jth entry and zero otherwise. For $q = 0, ..., h-1$, each $\boldsymbol{\Psi}_q$ is a $k \times k$ matrix, obtained from the vector moving average (VMA) model that is equivalent to the VECM (B1), namely:

$$\boldsymbol{r}_t = \boldsymbol{\Psi}(L)\boldsymbol{\epsilon}_t, \boldsymbol{\Psi}(L) = \sum_{q=0}^{\infty}\boldsymbol{\Psi}_q L^q \approx \sum_{q=0}^{h-1}\boldsymbol{\Psi}_q L^q,$$

where L is the lag operator. The net spillover from the information embedded in the ether spot price on exchange j to that on exchange i is calculated on each trading day as:

$$\theta_{ij}^* = \tilde{\theta}_{ij} - \tilde{\theta}_{ji}, \tilde{\theta}_{ij} = \frac{\theta_{ij}}{\sum_{s=1}^{k}\theta_{is}}.$$

Then θ_{ij}^* indicates whether market i is a "sender" or "receiver" of shocks compared with market j. Intuitively, if market i is a net sender of shocks relative to market j, as identified by $\theta_{ij}^* > 0$, then market i plays a more significant role in price discovery than market j (Diebold and Yilmaz 2012).

3. *Autocorrelation, independence, and state-persistence*: These tests are designed to reveal the level of market efficiency of the liquidity pools by looking at the autocorrelation, independence, and state-persistence of the return series. Under the efficient markets hypothesis, all valuable information in an efficient market should be fully, quickly, and accurately reflected in the current price. This means future prices in an efficient market should be unpredictable and follow a random walk. Rejecting this pattern (or equivalently, rejecting any of these tests) indicates a lack of efficiency of the market. For each exchange separately, we apply the following four tests to the entire time series of n minute-level ether spot returns, which is denoted $\mathbf{r} = \{r_1, ..., r_n\}$:

   1. Ljung and Box (1978) *test*: The test statistic takes the form:

$$n(n+2)\sum_{q=1}^{h}\left[r_q^2/(n-q)\right],$$

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where n is the sample size, r_q is the qth order autocorrelation coefficient estimated from $\mathbf{r}$ and h is the number of lags being tested, which we set at 1 because we want to check the short-term auto-correlation here; the longer term dependencies will be revealed by the Hurst exponent, defined below. The null hypothesis is that there is no autocorrelation present in the tested series.

2. *Wild-bootstrapped AVR test* (Andrews 1991; Choi 1999): This tests whether the returns are serially uncorrelated. To perform it we rely on the standardized variance ratio (VR) statistic of Choi (1999): For each sample $\mathbf{r}$ we set:

$$\mathrm{VR}(l) = \sqrt{2n/l}\sum_{q=1}^{n-1}\left(k\left(\frac{q}{l}\right)\frac{\sum_{t=1}^{n-q} r_t r_{t+q}}{\sum_{t=1}^{n} r_t^2}\right) \quad \text{and} \quad l = 1.3221 \frac{(4\gamma^2 n)^{0.2}}{(1-\gamma)^{0.8}},$$

where l is a lag truncation point,[24] γ is the lagged dependent variable coefficient estimate in an AR(1) model fitted to $\mathbf{r}$ and $k\left(\frac{q}{l}\right)$ is a quadratic spectral kernel that takes the form:

$$k\left(\frac{q}{l}\right) = 3\frac{\sin(y)/y - \cos(y)}{y^2},$$

with $y = 1.2\pi\left(\frac{q}{l}\right)$. Following Kim (2009), we obtain the p-value of the test in two steps. First, an optimal lag truncation point l^* is estimated according to equation (6.2) in Andrews (1991). The observed value of the test statistic is the corresponding VR value, denoted $\mathrm{VR}(l^*)$. Second, following Kim (2009), we take m bootstraps from the original sample $\mathbf{r}$ and in each bootstrapped sample we find the optimal lag truncation point, l_k^*, for $k = 1, ..., m$. Then we compute the corresponding $\mathrm{VR}\left(l_k^*\right)$ statistic for each k. The p-value of the test is computed as the proportion of bootstrapped statistics greater than the observed statistic.

3. Bartels (1982) *rank test of randomness*: The null hypothesis is that the sample is random. As a test of independence, this test is stronger than the previous two tests, which only rely on the autocorrelation coefficients of the returns. Denote a time series by $\{r_1, ..., r_n\}$. The test statistic is:

$$\frac{\mathrm{RVN} - 2}{\sigma},$$

where

$$\sigma^2 = \frac{4(n-2)(5n^2 - 2n - 9)}{5n(n+1)(n-1)^2},$$
$$\mathrm{RVN} = \frac{\sum_{i=1}^{n-1}(R_i - R_{i+1})^2}{\sum_{i=1}^{n}(R_i - (n+1)/2)^2},$$

and $R_i = \mathrm{rank}(r_i)$ for $i = 1, ..., n$.

4. *Hurst exponent* (Hurst 2021; 1951): This metric identifies the state persistence in a time series, which differs from autocorrelation. State persistence means the tendency for a system—in this case a token pair price—to remain at the same level. Reverse state persistence means the tendency for a token pair price not to remain at the same level. Positive (negative) autocorrelation is only a form of (reverse) state persistence. Primarily, autocorrelation identifies the recent dependencies within a series, whereas state persistence identifies long-term, extensive dependencies within the series. Calculating the R/S Hurst exponent relies on taking the entire time series $\mathbf{r} = \{r_1, ..., r_n\}$ and cutting this into sub-samples of equal size m, denoted $\{\mathbf{x}_k; k = 1, ..., n/m\}$. We demean all sub-samples by their sub-sample mean to obtain $\{\mathbf{y}_k\}$, and further convert them to their cumulative terms $\{\mathbf{z}_k\}$.[25] For each sub-sample k, we compute the "R/S" value as the ratio between the range size of $\mathbf{z}_k$ (i.e., $\max(\mathbf{z}_k) - \min(\mathbf{z}_k)$) and the standard deviation of $\mathbf{y}_k$. Taking all sub-samples, we thus obtain m different R/S values and subsequently compute their average R/S value, $\overline{\mathrm{R/S}}$.

   Now, we select a range of values for m, denoted by $\{m_i\}$. These values are chosen between 50 and 1440 and also have to be factors of the time series size n. For each m_i, we repeat the process discussed above and obtain an $\overline{\mathrm{R/S}}$ value, denoted by $\overline{\mathrm{R/S}}_i$. Then we run the following regression:

$$\ln(\overline{\mathrm{R/S}}_i) = a + b\ln(m_i) + e_i,$$

where b is called as the R/S Hurst exponent. A value of b near 0.5 indicates that the tested series $\mathbf{r}$ follows a Brownian motion. A value of b greater than 0.65 (less than 0.45) indicates significant (reverse) state persistence in $\mathbf{r}$.

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Appendix C

Supporting Empirical Results

TABLE C1 | Further preliminary statistics of data.

(a) Test stats of the Dickey-Fuller test *(H_0 : The series is nonstationary.)*

Log prices	**v3 (001)**	**v3 (005)**	**v3 (030)**	**v2**	**Bitstamp**	**Coinbase**
ETH-USDC vs ETH-USD		−1.327	−1.398	−1.393	−1.322	−1.317
ETH-USDC		−1.327	−1.398	−1.393	−1.46	
ETH-USDT		−1.33	−1.388	−1.354	−1.394	−1.315
BTC-ETH		−2.063	−2.105	−2.448	−2.111	−2.106
BTC-USDT			−0.617		−0.522	−0.458
BTC-USDC			−0.592		−0.496	
DAI-USDC	−73.407***	−52.050***				
ETH-DAI		−2.015	−1.402	−1.406		−1.352
ETH-LINK			−2.649	−3.093		−2.621
UNI-BTC			−2.272		−3.051	−2.844
LINK-USDT			−2.261			−1.711
Log returns						
vs ETH-USD		−158.288***	−155.508***	−182.077***	−158.805***	−158.379***
ETH-USDC		−158.288***	−155.508***	−182.077***	−165.910***	
ETH-USDT		−165.646***	−154.400***	−173.569***	−166.659***	−159.960***
BTC-ETH		−166.863***	−171.859***	−178.448***	−238.947***	−171.799***
BTC-USDT			−168.246***		−161.204***	−159.966***
BTC-USDC			−179.270***		−164.073***	
DAI-USDC	−206.645***	−171.335***				
ETH-DAI		−213.600***	−168.481***	−229.955***		−167.331***
ETH-LINK			−154.899***	−186.140***		−165.333***
UNI-BTC			−161.178***		−227.954***	−158.532***
LINK-USDT			−417.993***			−147.820***

(b) Panel B: Sample level test stats of Johansen test and % of days in which H_0 is rejected *(H_0 : The system of log prices is cointegrated at a rank of r.)*

		$r = 0$	$r = 1$	$r = 2$	$r = 3$	$r = 4$
ETH-USDC vs ETH-USD	Test stats.	717,980.8***	362,408.3***	132,869.9***	61,612.8***	1.7
	Rejection %	100.00	100.00	99.62	92.87	11.54
ETH-USDC	Test stats.	325,683.5***	95,964.8***	27,204.7***	1.5	
	Rejection %	99.34	96.44	73.55	9.38	
ETH-USDT	Test stats.	736,884.4***	335,619.4***	147,850.6***	55,838.3***	1.7
	Rejection %	100.00	100.00	99.72	94.56	12.48
BTC-ETH	Test stats.	327,409.7***	81,283.6***	20,248.7***	7631.2***	4.7
	Rejection %	100.00	98.53	65.91	26.92	4.81
BTC-USDT	Test stats.	86,863.1***	24,767.0***	0.3		
	Rejection %	99.44	76.34	12.58		
BTC-USDC	Test stats.	32,615.1***	0.2			
	Rejection %	83.96	11.54			

(Continues)

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TABLE C1 | (Continued)

(b) Panel B: Sample level test stats of Johansen test and % of days in which H_0 is rejected *(H_0 : The system of log prices is cointegrated at a rank of r.)*

		$r = 0$	$r = 1$	$r = 2$	$r = 3$	$r = 4$
DAI-USDC	Test stats.	37,787.7***	9260.9***			
	Rejection %	94.20	46.54			
ETH-DAI	Test stats.	247,495.0***	134,643.0***	34,833.7***	1.7	
	Rejection %	98.59	90.89	72.86	9.30	
ETH-LINK	Test stats.	75,018.9***	21,030.9***	6.6		
	Rejection %	96.72	78.05	10.41		
UNI-BTC	Test stats.	15,948.2***	25.2***	7.4***		
	Rejection %	56.70	14.71	2.12		
LINK-USDT	Test stats.	1846.9***	2.9			
	Rejection %	23.10	5.07			

Note: Our sampling period is between July 1, 2021 to June 1, 2024. Panel A presents the results of Dickey-Fuller test for the minute-level ether log price and returns series from the four exchanges. The reported test statistics of the Dickey-Fuller test is MacKinnon's approximate statistics based on Levin et al. (2002). Panel B presents the results of Johansen test using the minute-level log prices in the entire sample (the first row) and in each trading day across our sampling period at 1% significance level. *, **, and *** represents rejecting the null hypothesis at 10%, 5%, and 1% level, respectively.

TABLE C2 | Gross and net spillover (in %).

(a) BTC, ETH vs USD.

	From	v3 (005)	v3 (030)	v2	Bitstamp	Coinbase
	To					
ETH-USDC	v3 (005)	0.545	0.012	0.059	0.171	0.172
vs ETH-USD	v3 (030)	0.007	0.942	0.015	0.004	0.005
	v2	0.035	0.016	0.818	0.023	0.023
	Bitstamp	0.199	0.01	0.047	0.453	0.34
	Coinbase	0.199	0.009	0.045	0.338	0.45
	net spillover	−0.025	−0.015	−0.055	0.056	0.048
ETH-USDC	v3 (005)	0.889	0.011	0.06	0.021	
	v3 (030)	0.013	0.955	0.017	0.005	
	v2	0.067	0.016	0.88	0.017	
	Bitstamp	0.018	0.003	0.011	0.96	
	net spillover	0	−0.004	−0.001	0.006	
ETH-USDT	v3 (005)	0.774	0.007	0.025	0.035	0.136
	v3 (030)	0.009	0.943	0.017	0.009	0.011
	v2	0.032	0.018	0.86	0.025	0.041
	Bitstamp	0.036	0.006	0.017	0.857	0.066
	Coinbase	0.138	0.01	0.032	0.07	0.731
	net spillover	0.019	−0.01	−0.036	0.013	0.015
BTC-ETH	v3 (005)	0.949	0.004	0.002	0.012	0.016
	v3 (030)	0.004	0.977	0.003	0.003	0.004
	v2	0.001	0.002	0.986	0.002	0.002
	Bitstamp	0.011	0.003	0.003	0.771	0.189
	Coinbase	0.015	0.003	0.003	0.191	0.761
	net spillover	−0.01	−0.001	0	0.007	0.003

(Continues)

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TABLE C2 | (Continued)

(a) BTC, ETH vs USD.						
	From	**v3 (005)**	**v3 (030)**	**v2**	**Bitstamp**	**Coinbase**
BTC-USDT	v3 (030)		0.991		0.004	0.004
	Bitstamp		0.004		0.871	0.115
	Coinbase		0.004		0.133	0.854
	net spillover		0		0.013	−0.012
BTC-USDC	v3 (030)		0.995		0.005	
	Bitstamp		0.005		0.995	
	net spillover		0		0	

(b) Other tokens.							
	From	**v3 (001)**	**v3 (005)**	**v3 (030)**	**v2**	**Bitstamp**	**Coinbase**
	To						
DAI-USDC	v3 (001)	1	0				
	v3 (005)	0	1				
	net spillover	0	0				
ETH-DAI	v3 (005)		0.966	0.002	0.005		0.018
	v3 (030)		0.003	0.976	0.009		0.005
	v2		0.005	0.009	0.966		0.007
	Coinbase		0.015	0.005	0.008		0.954
	net spillover		0	0	−0.001		0
ETH-LINK	v3 (030)			0.977	0.007		0.008
	v2			0.007	0.987		0.002
	Coinbase			0.009	0.003		0.986
	net spillover			0	0		−0.001
UNI-BTC	v3 (030)			0.998		0	0.002
	Bitstamp			0		0.996	0.003
	Coinbase			0.002		0.004	0.991
	net spillover			0		0	0
LINK-USDT	v3 (030)			1			0
	Coinbase			0			1
	net spillover			0			0

TABLE C3 | Panel A documents minimum, mean, median, and maximum value of correlations between explanatory variables include trading volume (TV), large trade ratio (LTR), market depth (λ), liquidity providers activeness (LPA), volatility (Vol), and basis, as well as price discovery ability and efficiency measures, CS, MIS, and ADH, across all pairs on all exchanges.

(a) Min, mean, median, and max values of correlations between explanatory variables.						
		LTR	λ	**LPA**	**Vol**	**Basis**
TV	Min	−0.415	−0.26	−0.011	−0.009	−0.172
	Median	0.102	−0.054	0.097	0.442	0.295
	Mean	0.015	−0.065	0.143	0.404	0.252
	Max	0.364	0.093	0.671	0.683	0.632
LTR	Min		−0.256	−0.038	−0.481	−0.456

(Continues)

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TABLE C3 | (Continued)

(a) Min, mean, median, and max values of correlations between explanatory variables.

		LTR	λ	LPA	Vol	Basis
	Median		0.009	0.041	0.105	−0.06
	Mean		0.003	0.064	0.035	−0.098
	Max		0.198	0.251	0.399	0.183
λ	Min			−0.168	−0.061	−0.125
	Median			−0.012	0.074	0.068
	Mean			−0.02	0.193	0.119
	Max			0.109	0.683	0.535
LPA	Min				−0.191	−0.302
	Median				0.033	0
	Mean				0.073	0.005
	Max				0.288	0.192
Vol	Min					−0.236
	Median					0.274
	Mean					0.285
	Max					0.597

(b) Vol-basis correlations, sorted from the highest to the lowest.

Coin pair	Exchange	Correlation
LINK-USDT	v3 (030)	0.597
LINK-ETH	v2	0.590
BTC-ETH	v3 (030)	0.422
ETH-DAI	v3 (005)	0.402
LINK-ETH	v3 (030)	0.388
ETH-USDC	v3 (005)	0.364
USDT-ETH	v3 (005)	0.363
BTC-USDT	v3 (030)	0.317
ETH-USDC	v2	0.302
ETH-USDC	v3 (030)	0.296
BTC-USDC	v3 (030)	0.274
BTC-ETH	v2	0.252
USDC/USD-ETH	v2	0.234
USDT-ETH	v3 (030)	0.231
ETH-DAI	v3 (030)	0.227
USDC/USD-ETH	v3 (030)	0.216
BTC-ETH	v3 (005)	0.206
USDT-ETH	v2	0.205
USDC/USD-ETH	v3 (005)	0.171
ETH-DAI	v2	0.165
UNI-BTC	v3 (030)	−0.236

Note: Panel B documents correlations between Vol and the basis for each coin pair traded in every exchange.

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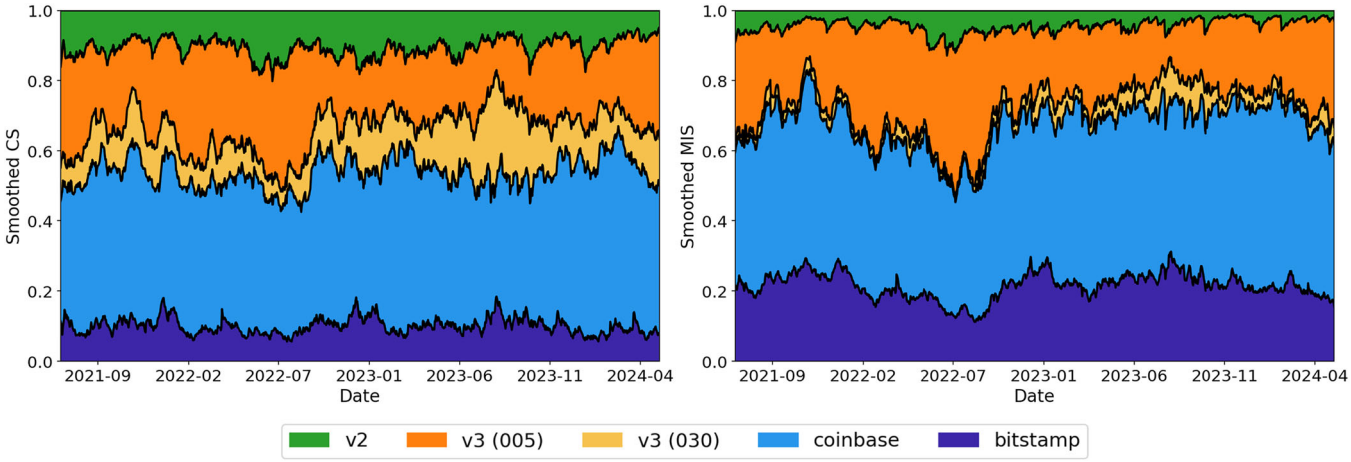

FIGURE C1 | CS and MIS value (smoothed) plots on most traded pair ETH-USDC versus ETH-USD.

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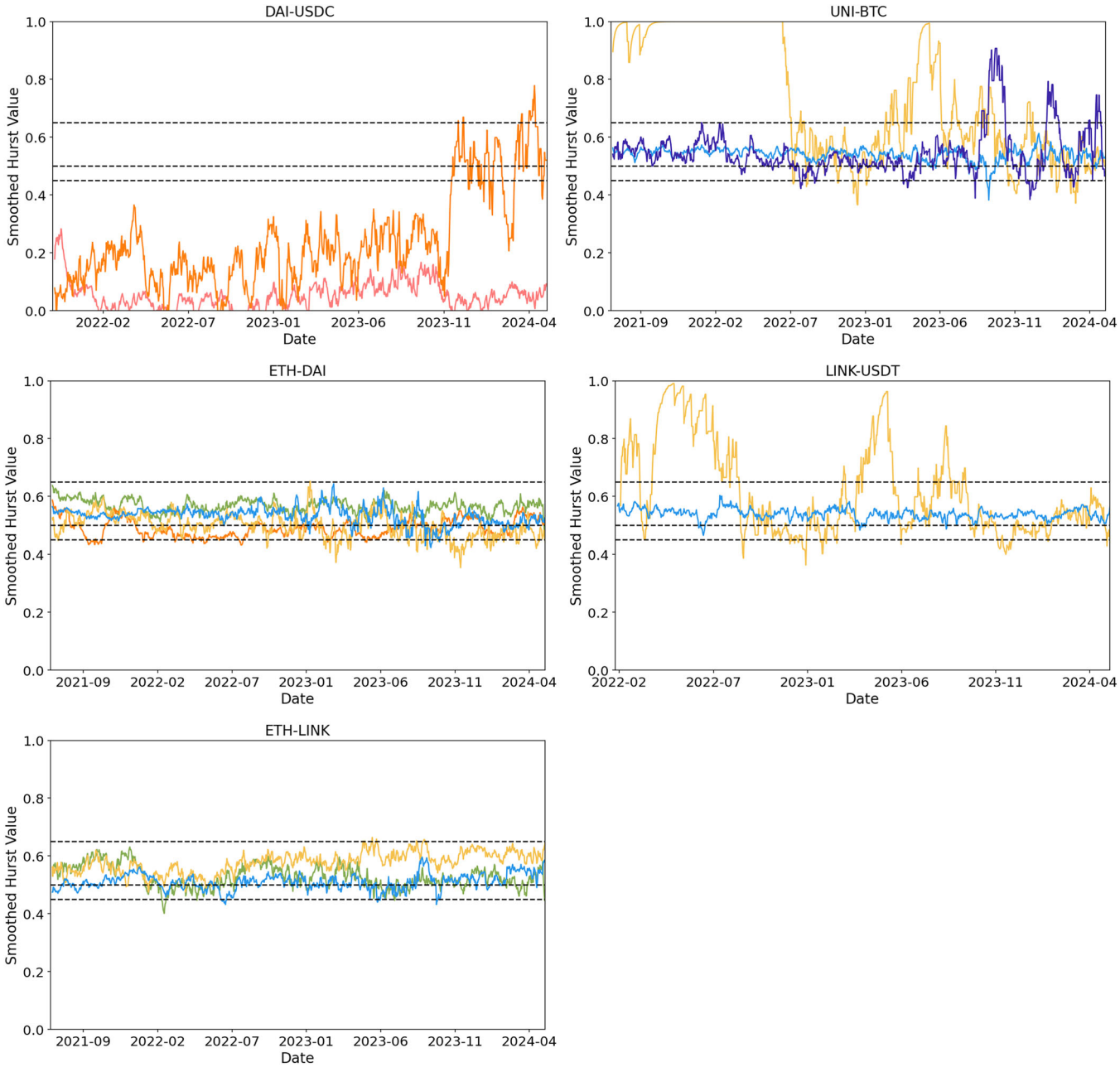

FIGURE C2 | R/S Hurst exponents (smoothed) for minor pools.

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TABLE C4 | The impact of market characteristics on price discovery and market efficiency.

Coin Pair	CS					MIS					ADH				
	TV	Basis	LTR	λ	LPA	TV	Basis	LTR	λ	LPA	TV	Basis	LTR	λ	LPA
v3 (001)															
DAI-USDC	0.067*		0.148***	0.026	0.061*	0.081**		0.137***	0.019	0.068**	0.005		−0.074**	0.161***	0.065*
v3 (005)															
USDC/ USD-ETH	0.296***	−0.092***	−0.145***	0.126***	−0.03	0.375***	−0.079***	−0.112***	0.273***	−0.019	−0.069*	0.057*	−0.033	−0.207***	−0.043
ETH-USDC	−0.014	0.035	−0.103***	−0.036	−0.045	−0.094**	0.061	−0.097***	−0.049	−0.046	0.014	−0.132***	−0.068**	−0.231***	−0.044
USDT-ETH	0.237***	−0.267***	−0.095***	−0.135***	0.139***	0.268***	−0.340***	−0.134***	−0.107***	0.143***	0.094***	−0.114***	0.017	−0.103***	0.175***
BTC-ETH	0.190***	0.139***	−0.079**	−0.190***	−0.002	0.242***	0.167***	−0.038	−0.103***	0.096***	−0.056*	0.034	0.025	0.035	−0.03
ETH-DAI	−0.049*	−0.133***	−0.478***	−0.021	0.072***	−0.167***	−0.130***	−0.388***	−0.024	0.023	−0.044	0.045	0.029	0.010	0.063**
DAI-USDC	−0.068**		−0.277***	−0.206***	−0.086***	−0.061*		−0.253***	−0.176***	−0.093***	−0.003		0.078	0.146***	0.078**
v3 (030)															
USDC/ USD-ETH	−0.209***	−0.039	0.139***	−0.073**	0.011	−0.128***	−0.030	0.109***	−0.050	−0.006	−0.071**	−0.033	0.009	0.016	0.014
ETH-USDC	−0.145***	−0.080**	0.107***	−0.049	0.014	−0.080**	−0.096**	0.080**	−0.032	−0.007	−0.051	−0.084**	−0.009	0.013	0.012
USDT-ETH	−0.105***	−0.094**	0.121***	−0.040	0.027	−0.033	−0.135***	0.062*	−0.023	0.051*	0.011	−0.051	0.117***	−0.060**	−0.03
BTC-ETH	−0.269***	−0.109***	−0.039	−0.002	−0.064**	−0.263***	0.153***	−0.021	0.037	−0.059**	−0.148***	−0.071*	−0.079**	−0.164***	−0.04
ETH-DAI	−0.071**	−0.113***	0.108***	0.093***	−0.022	−0.090***	−0.108***	0.098***	0.179***	0.008	−0.109***	−0.033	−0.031	−0.038	−0.004
BTC-USDT	−0.162***	−0.188***	0.177***	−0.051*	0.074*	−0.135***	−0.128***	0.234***	−0.033	0.049	−0.084*	−0.022	−0.043	−0.020	0.02
BTC-USDC	−0.268***	−0.327***	0.088***	0.024	−0.015	−0.222***	−0.275***	0.103***	0.008	−0.018	−0.114***	−0.062*	0.004	−0.047	−0.014
LINK-ETH	−0.014	−0.290***	0.165***	−0.166***	0.003	−0.005	−0.229***	0.195***	−0.139***	0.037	−0.011	−0.118***	−0.054*	−0.032	−0.008
LINK-USDT	−0.042	−0.124***	−0.238***	−0.067*	−0.014	−0.034	−0.037	−0.207***	−0.036	0	−0.005	0.043	−0.039	0.007	−0.024
UNI-BTC	−0.174***	0.430	−0.138***	−0.028	0.018	−0.132***	0.951***	−0.138***	0.067	0.011	−0.038	2.224*	−0.064	−0.094	0.006
v2															
USDC/ USD-ETH	0.040	−0.119***	0.045	−0.037	−0.036	0.132***	−0.107***	0.166***	−0.017	−0.03	−0.094***	0.076**	0.105***	−0.048	0.001
ETH-USDC	0.276***	−0.124***	0.161***	−0.031	−0.059*	0.382***	−0.147***	0.250***	−0.013	−0.051*	−0.128***	0.113***	0.103***	−0.048	0.006
USDT-ETH	−0.019	−0.090**	0.091***	−0.034	0.003	0.200***	−0.227***	0.110***	0.029	0.011	−0.200***	0.202***	0.180***	0.105***	0.018
BTC-ETH	−0.160***	−0.081**	−0.117***	−0.083**	0.026	−0.176***	−0.025	−0.013	0.075**	−0.003	−0.111***	−0.017	0.011	−0.132**	−0.002
ETH-DAI	−0.115***	−0.149***	−0.047	−0.035	−0.01	−0.122***	−0.123***	−0.01	−0.016	−0.007	−0.042	0.035	0.059*	0.008	−0.005
LINK-ETH	0.081**	−0.229***	0.008	−0.053	−0.021	−0.059*	0.042	0.054	−0.034	−0.023	0.032	−0.040	−0.022	0.073**	0.029

Note: The results of model (5), which uses trading volume (TV), basis, large trade ratio (LTR), the market illiquidity measure λ in (3), and the liquidity provider activity (LPA) to explain price discovery ability (CS or MIS) and efficiency (ADH). The sample from 1 July 2021 and 1 June 2024, 1066 observations. All variables are normalized and asterisks ***, **, and * represent 1%, 5%, and 10% significance levels.

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